

DIETARY FIBRE AIDS IN THE MANAGEMENT OF CAT AND DOG GASTROINTESTINAL DISEASE

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Veterinary Medical Center





When Should I be Thinking about Fiber in My Gastroenterology Cases?



Main GI Clinical Indications

- Acute Diarrhea
- Inflammatory Bowel Disease (Colitis)
- Irritable Bowel Syndrome
- Constipation

Acute Diarrhea



Infectious Diarrhea Differentials



Organism	Organ	Evidence
Helminths		
Whipworms	Colon	Good
Hookworms	SI, Colon	Good
<i>Heterobilharzia americana</i>	Colon	Good
Protozoa		
<i>Balantidium coli</i>	Colon	Weak
<i>Cryptosporidium parvum</i>	SI, Colon	Mod-Good
<i>Entamoeba histolytica</i>	Colon	Poor
<i>Giardia canis</i>	SI, Colon	Mod-Good
<i>Isospora canis</i>	Colon	Weak
<i>Toxoplasma gondii</i>	SI, Colon	Good
<i>Tritrichomonas foetus</i>	SI, Colon	Mod-Good
Fungi		
<i>Histoplasma capsulatum</i>	SI, Colon	Good

Organism	Organ	Evidence
<i>Pythium insidiosum</i>	Stomach, SI, Colon	Good
Algae		
<i>Prototheca zopfii</i>	Colon	Good
Bacteria		
<i>Campylobacter spp.</i>	SI, Colon	Moderate
<i>Clostridium spp.</i>	Colon	Weak-Mod
<i>E. coli spp.</i>	SI, Colon	Good
<i>Salmonella spp.</i>	SI, Colon	Good
<i>Brachyspira spp.</i>	SI, Colon	Weak-Mod



Acute Studies

First study included dogs fed the high fiber diet had a higher proportion of improved fecal scores compared to dogs fed the standard diet in the short-term.

▫Concurrent use of Metro

Second study evaluated dogs with noninfectious acute colitis randomized into three placebo-controlled groups

▫Easily digestible diet in combination with placebo

▫Easily digestible diet in combination with metronidazole

▫Psyllium-enhanced easily digestible diet in combination with placebo

Abstracts in puppies and kittens in shelters with diarrhea

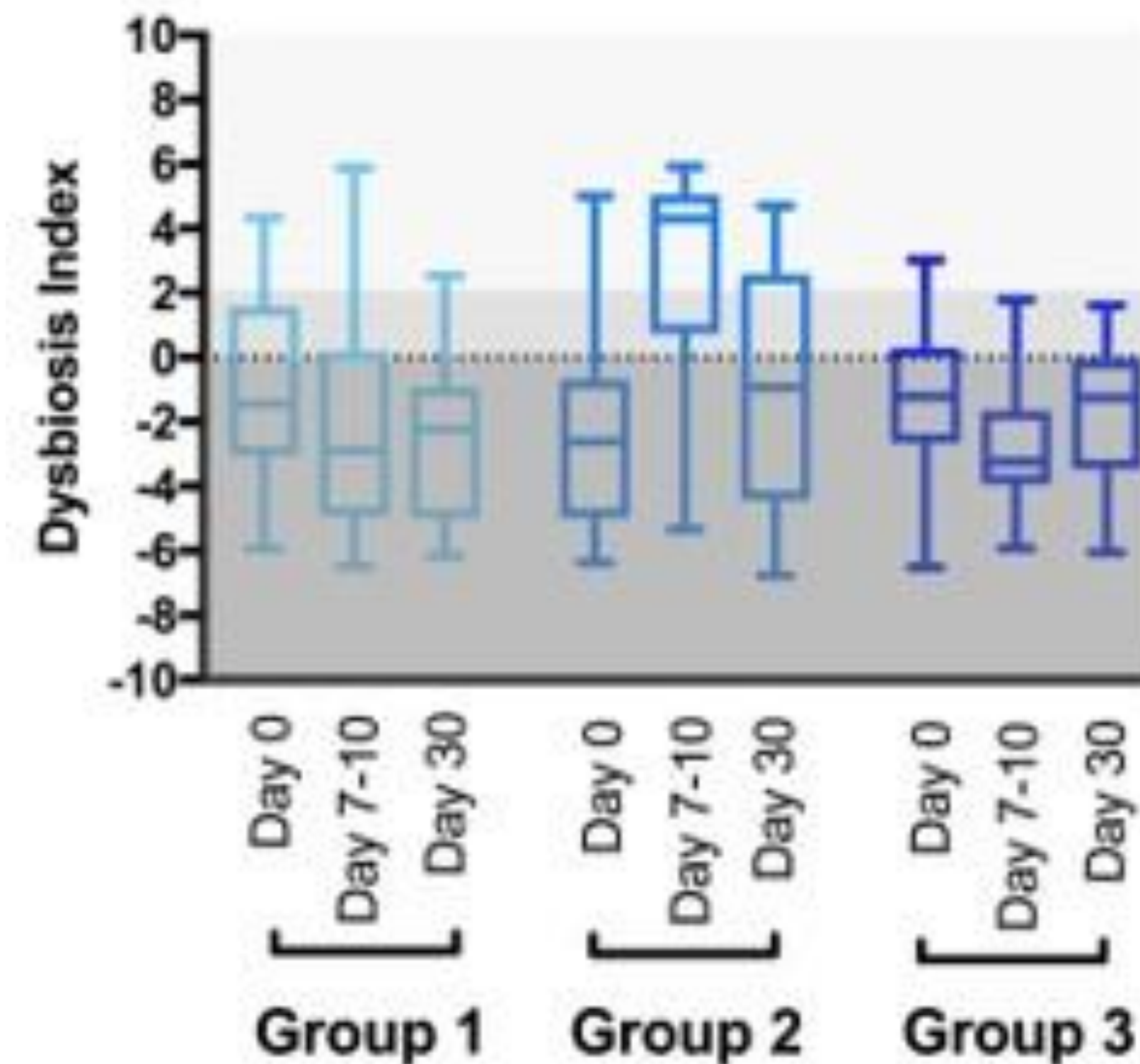
Rudinsky AJ, et al. 2022.

Lappin MR, et al. 2019.

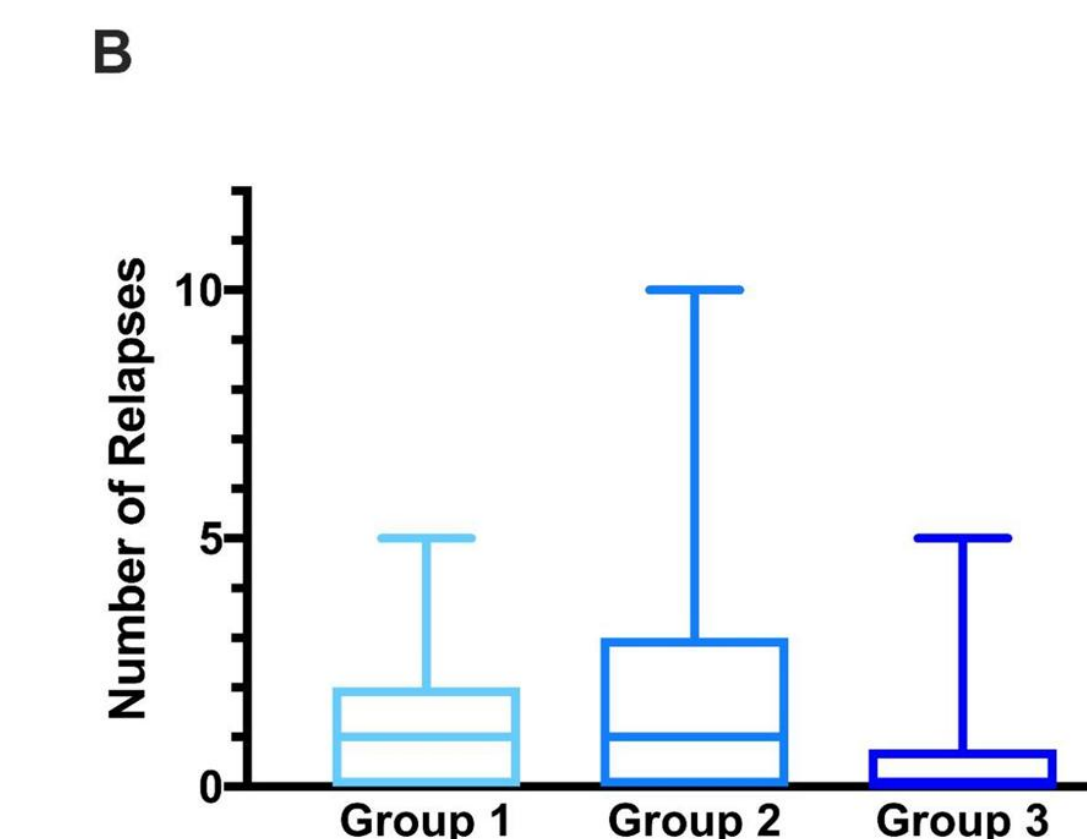
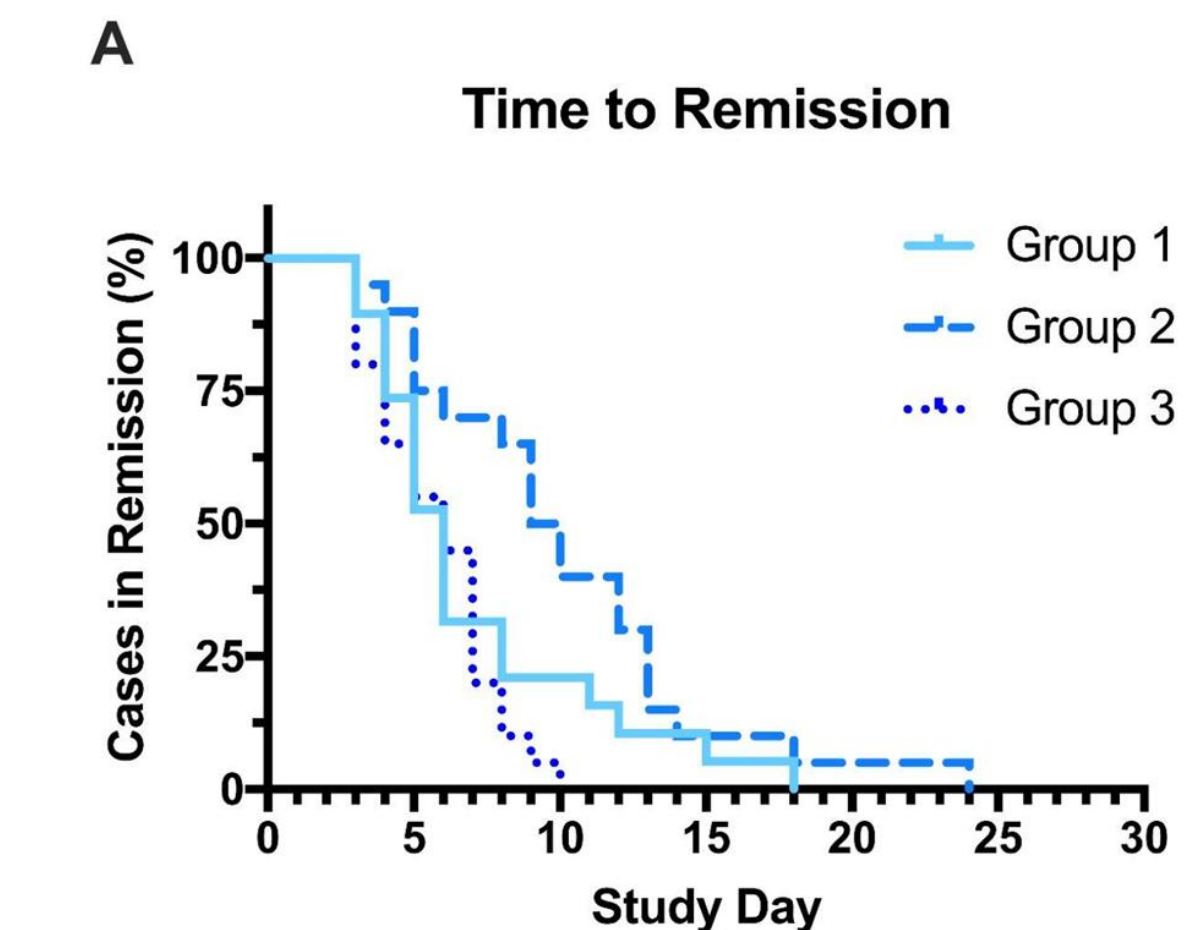
Franz NZ, et al. 2020.

Franz NZ, et al. 2020.

Fibre for the Win!



Rudinsky AJ, et al. 2022.



Choosing a Diet (Acute)!

Easily Digestible



Fibre Enhanced



KEY POINT

Current studies provide evidence for the use of fiber-enhanced diets for the management of acute colitis in dogs and cats with improved outcomes compared to traditional empirical management options.

Rudinsky AJ, et al. 2022.
Lappin MR, et al. 2019.
Franz NZ, et al. 2020.
Franz NZ, et al. 2020.



A Randomized Double Blinded Placebo-Controlled Clinical Trial of a Probiotic or Metronidazole for Acute Canine Diarrhea

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TABLE 1 | Comparison of patient characteristics, history, additional treatments, laboratory testing, and time to formed feces between study treatment groups.

	Probiotic (n = 20)	Metronidazole (n = 20)	Placebo (n = 20)	p-value ^{††}	All Treatment Groups (n = 60)
Time to formed feces (days)	3.5 ± 2.2	4.6 ± 2.4	4.8 ± 2.9	0.17	
Age (yrs)	5.3 ± 3.2	5.7 ± 3.7	5.7 ± 3.9	0.94	5.6 ± 3.6
Body weight (kg)	20.2 ± 11.3	22.8 ± 13.0	20.9 ± 11.6	0.78	21.2 ± 11.8
Duration before presentation (days)	1.7 ± 1.4	1.8 ± 2.3	1.6 ± 1.4	0.94	1.7 ± 1.4
Previous history of diarrhea (n)	10	11	13	ns	34
Dietary indiscretion (n)	9	4	8	ns	21
Stress (n)	3	4	2	ns	9
Food change (n)	1	3	5	ns	9
Vomiting (n)	10	10	8	ns	28
Hematochezia (n)	14	13	12	ns	39
Fluid therapy (n)	8	6	9	ns	23
Maropitant (n)	11	11	8	ns	30
Fenbendazole (n)	1	2	3	ns	6
Hematocrit (%) (ref: 37.3–61.7)	50.2 ± 5.4	49.7 ± 6.9	47.4 ± 8.3	0.42	49.1 ± 7.0
Total protein (g/dL) (ref: 5.2–8.2)	6.7 ± 0.5	6.8 ± 0.7	6.6 ± 0.6	0.71	6.7 ± 0.6
Creatinine (mg/dL) (ref: 0.5–1.8)	1.0 ± 0.3	1.1 ± 0.3	1.1 ± 0.3	0.25	1.1 ± 0.3
Sodium (mEq/L) (ref: 142–160)	149 ± 3.9	151 ± 2.9	150 ± 2.5	0.14	150 ± 3.2
Potassium (mEq/L) (ref: 3.5–5.8)	4.1 ± 0.6	4.5 ± 0.7	4.3 ± 0.5	0.12	4.3 ± 0.6
Chloride (mEq/L) (ref: 107–122)	114 ± 3.7	115 ± 3.4	115 ± 3.5	0.59	115 ± 3.5
Toxocara canis ova [†]	0/17	2/15	2/17	ns	4/49
CPA positive [†]	8/9	10/11	13/15	ns	31/35
CPE positive [†]	3/9	5/11	8/15	ns	16/35
CPnetEF positive [†]	2/9	4/11	6/15	ns	12/35
Campylobacter coli positive [†]	0/9	0/11	1/15	ns	1/35
Circovirus positive [†]	0/9	0/11	2/15	ns	2/35
Episode of diarrhea after study (n)	4	1	3	0.48	8

[†] numbers given are the n positive/n samples submitted. Not all dogs had sufficient sample for analysis. ^{††} No results were statistically significant; ns indicates there were no significant differences for pairwise two-proportion z-tests which were assessed using Bonferroni correction (p < 0.017 considered significant). CPA, C. perfringens alpha toxin gene; CPE, Clostridium perfringens enterotoxin (CPE); CPnetEF, Clostridium perfringens netEF gene (CPnetEF).



Metronidazole treatment of acute diarrhea in dogs: A randomized double blinded placebo-controlled clinical trial

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Funding information

Zomedica, Inc.

Abstract

Background: Metronidazole is commonly administered to dogs with acute diarrhea, but there is limited evidence to support this practice.

Objective: To investigate the effects of metronidazole administration on dogs with acute nonspecific diarrhea.

Animals: Thirty-one dogs, including 14 test population dogs and 17 controls.

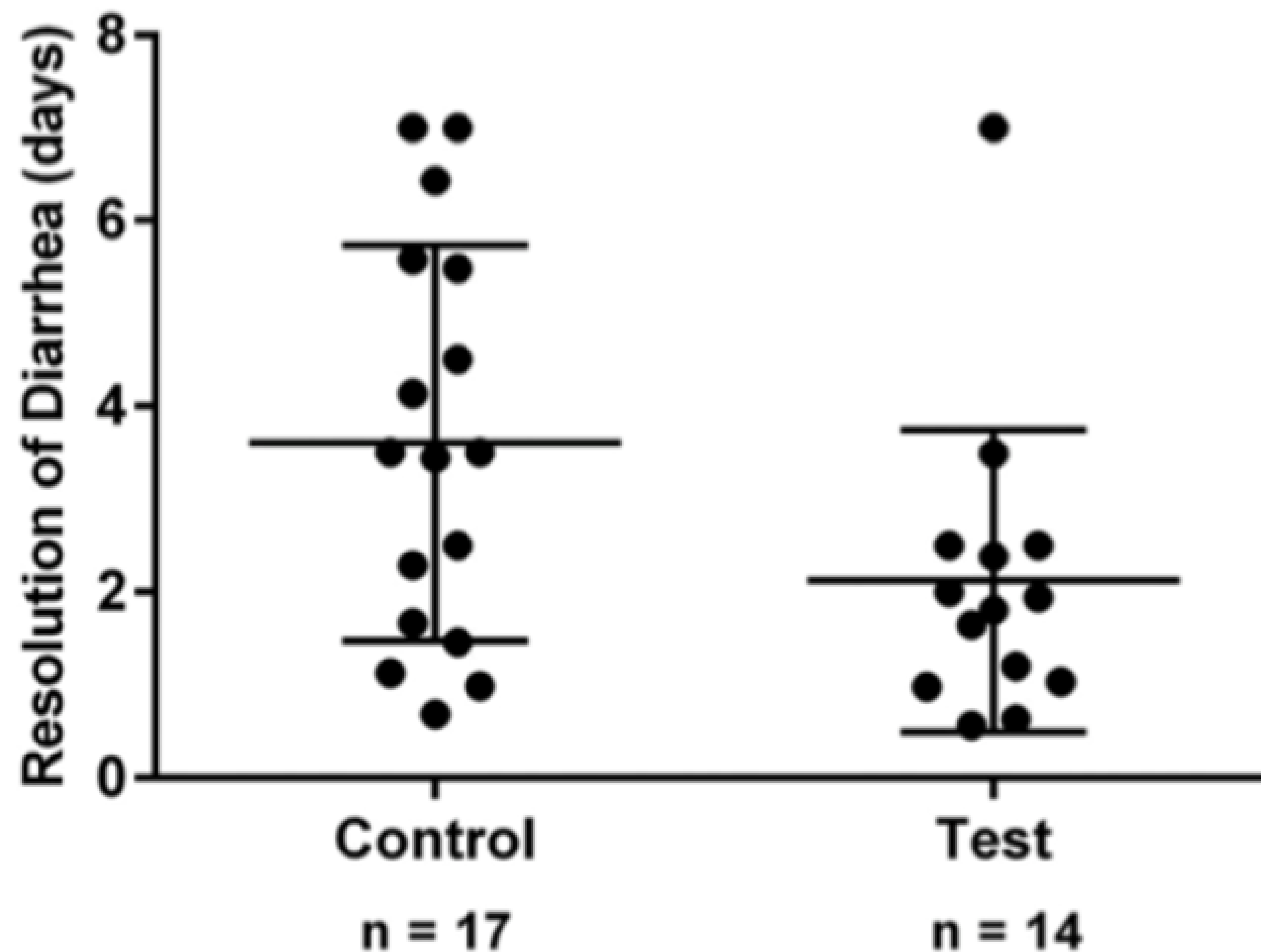
Methods: Randomized controlled clinical trial. Dogs with acute diarrhea in which causation was not determined by routine fecal diagnostic testing were randomly assigned to metronidazole treatment (10–15 mg/kg PO q12h for 7 days) or placebo. Fecal cultures and characterization of *Clostridium perfringens* isolates also were performed. Owners maintained medication and fecal scoring logs, and fecal diagnostic tests were repeated on day 7.

Results: The mean $\pm$ SD time to resolution of diarrhea for test population dogs (2.1 ± 1.6 days) was less than that for controls (3.6 ± 2.1 days, $P = .04$). Potential relationships of *C. perfringens* with acute diarrhea pathogenesis were not investigated, but only 3 of 13 (23.1%) test population dogs had persistent *C. perfringens* carriage at day 7, which was less than the 11 of 14 (78.6%) controls with persistent growth ($P = .007$).

Conclusions and Clinical Importance: Our results suggest that metronidazole treatment can shorten duration of diarrhea and decrease fecal culture detection of *C. perfringens* in some dogs with acute nonspecific diarrhea. Additional studies are needed to assess the benefits and risks of routine use of metronidazole for this purpose because most dogs achieve resolution of diarrhea within several days regardless of treatment.

KEYWORDS

Clostridium perfringens, enterotoxin, gastroenteritis, therapy



Metronidazole treatment of acute diarrhea in dogs: A randomized double blinded placebo-controlled clinical trial

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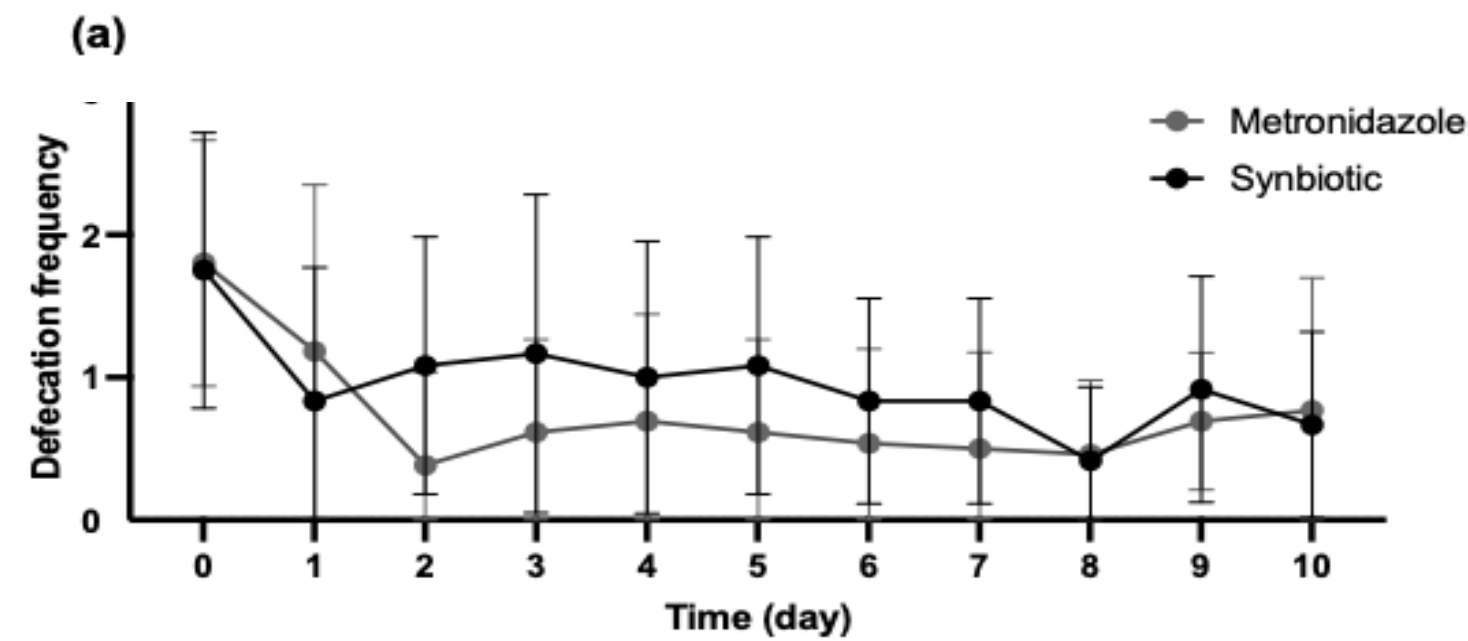
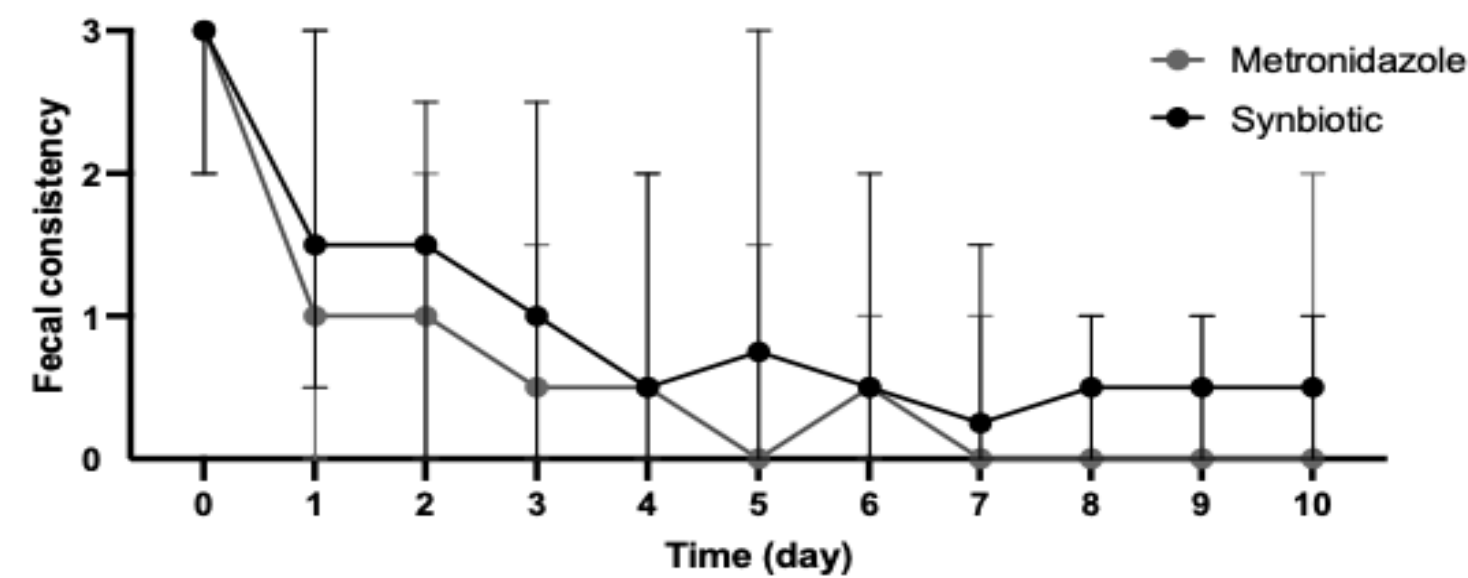
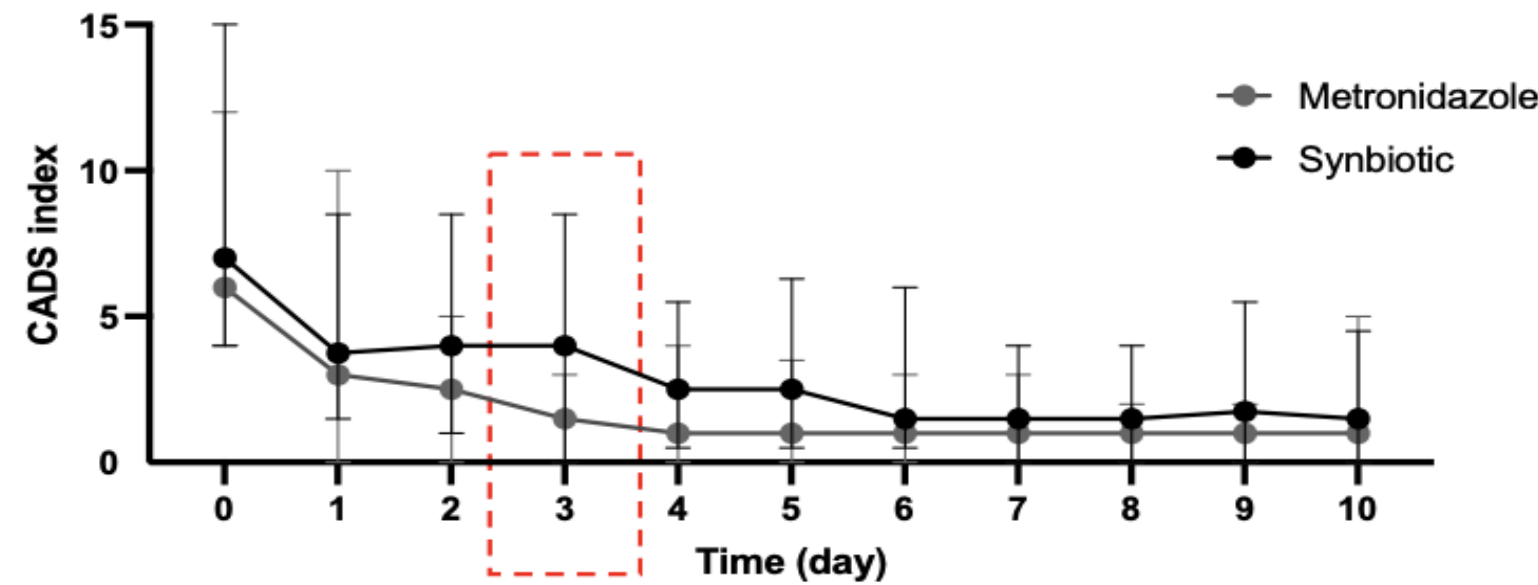
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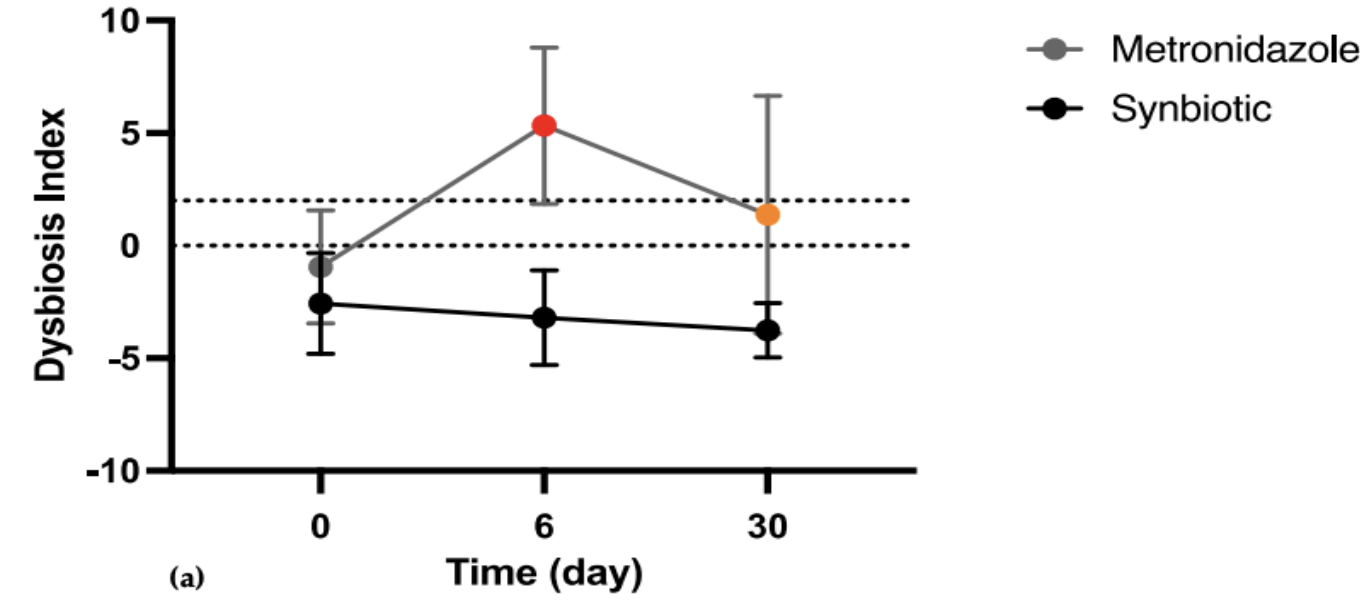
Article

The Effect of Metronidazole versus a Synbiotic on Clinical Course and Core Intestinal Microbiota in Dogs with Acute Diarrhea

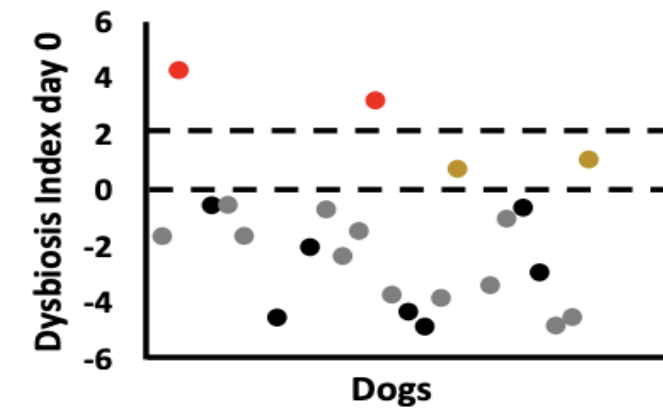
Helene Stübing ^{1,*}, Jan S. Suchodolski ², Andrea Reisinger ¹, Melanie Werner ³, Katrin Hartmann ¹ , Stefan Unterer ³ and Kathrin Busch ¹ 



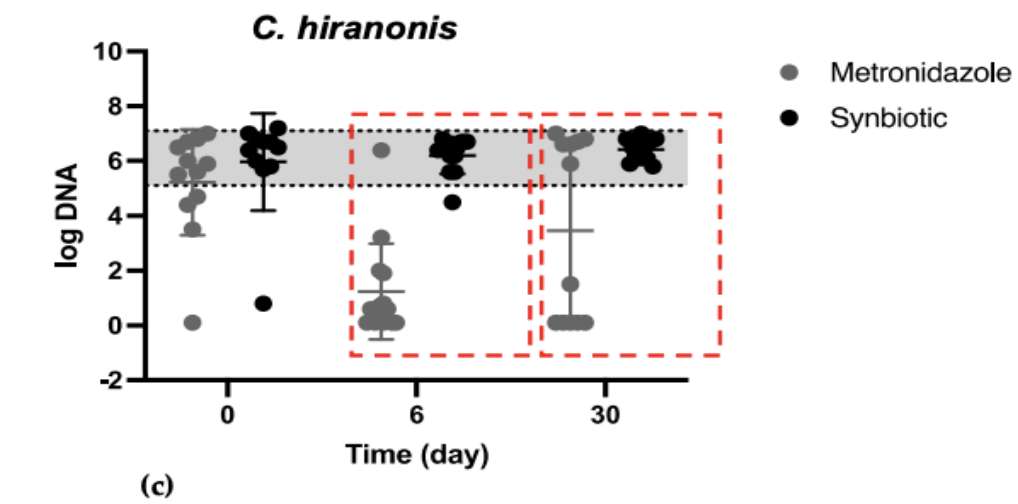
(b)



(a)



(b)



(c)

RESEARCH ARTICLE

Target trial emulation: Do antimicrobials or gastrointestinal nutraceuticals prescribed at first presentation for acute diarrhoea cause a better clinical outcome in dogs under primary veterinary care in the UK?

Camilla Pegram ^{1,*}, Karla Diaz-Ordaz ², Dave C. Brodbelt ¹, Yu-Mei Chang ³, Sarah Tayler ⁴, Fergus Allerton ⁵, Lauren Prisk ¹, David B. Church ⁴, Dan G. O'Neill ¹ 

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OPEN ACCESS

Citation: Pegram C, Diaz-Ordaz K, Brodbelt DC, Chang Y-M, Tayler S, Allerton F, et al. (2023) Target trial emulation: Do antimicrobials or gastrointestinal nutraceuticals prescribed at first presentation for acute diarrhoea cause a better clinical outcome in dogs under primary veterinary care in the UK? PLoS ONE 18(10): e0291057. <https://doi.org/10.1371/journal.pone.0291057>

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Data Availability Statement: Data relevant to this study are available from figshare at DOI: [10.6084/m9.figshare.24065352](https://doi.org/10.6084/m9.figshare.24065352).

Funding: CP is supported at the RVC by an award from the Dogs Trust Canine Welfare Grants (number 5654). URL: <https://www.dogstrust.org.uk/how-we-help/the-future/research> The funders reviewed the manuscript and were involved in the

Abstract

Target trial emulation applies design principles from randomised controlled trials to the analysis of observational data for causal inference and is increasingly used within human epidemiology. Veterinary electronic clinical records represent a potentially valuable source of information to estimate real-world causal effects for companion animal species. This study employed the target trial framework to evaluate the usefulness on veterinary observational data. Acute diarrhoea in dogs was used as a clinical exemplar. Inclusion required dogs aged ≥ 3 months and < 10 years, presenting for veterinary primary care with acute diarrhoea during 2019. Treatment strategies were: 1. antimicrobial prescription compared to no antimicrobial prescription and 2. gastrointestinal nutraceutical prescription compared to no gastrointestinal nutraceutical prescription. The primary outcome was clinical resolution (defined as no revisit with ongoing diarrhoea within 30 days from the date of first presentation). Informed from a directed acyclic graph, data on the following covariates were collected: age, breed, bodyweight, insurance status, comorbidities, vomiting, reduced appetite, haematochezia, pyrexia, duration, additional treatment prescription and veterinary group. Inverse probability of treatment weighting was used to balance covariates between the treatment groups for each of the two target trials. The risk difference (RD) of 0.4% (95% CI -4.5% to 5.3%) was non-significant for clinical resolution in dogs treated with antimicrobials compared with dogs not treated with antimicrobials. The risk difference (RD) of 0.3% (95% CI -4.5% to 5.0%) was non-significant for clinical resolution in dogs treated with gastrointestinal nutraceuticals compared with dogs not treated with gastrointestinal nutraceuticals. This study successfully applied the target trial framework to veterinary observational data. The findings show that antimicrobial or gastrointestinal prescription at first presentation of acute



Comparison of the Therapeutic Effect of Treatment with Antibiotics or Nutraceuticals on Clinical Activity and the Fecal Microbiome of Dogs with Acute Diarrhea

Giulia Pignataro ^{1,*}, Roberta Di Prinzio ¹, Paolo Emidio Crisi ^{1,*}, Benedetta Belà ¹, Isa Fusaro ¹, Carlo Trevisan ², Luigi De Acetis ³ and Alessandro Gramenzi ¹

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² Veterinary Practice, “Ambulatorio Veterinario Dr. Carlo Trevisan”, 66016 Guardiagrele, CH, Italy; gledd@hotmail.it
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* Correspondence: gpignataro@unite.it (G.P.); pocrisi@unite.it (P.E.C.)

Simple Summary: Acute diarrhea in dogs is one of the most common reasons for veterinary visits. Although this disorder is generally self-limiting, antibiotics are still frequently used as treatment for acute diarrhea in clinical practice. Antimicrobial resistance represents a major challenge for public health and requires immediate and drastic solutions. To date, the emergence and spread of antimicrobial resistance has been attributed to the misuse or indiscriminate use of antibiotics. The aim of this study is to compare the effects on clinical activity and fecal microbiota of the administration of an antibiotic combination in comparison to a nutraceutical product in dogs with acute non-hemorrhagic diarrhea. The results of the present study suggest that this nutraceutical treatment had a similar clinical effect compared to the antibiotic formulation and may represent an alternative to commonly used antimicrobial therapy.

Abstract: Dogs with acute diarrhea are often presented to clinical practice and, although this generally represents a self-limiting condition, antibiotics are still frequently used as treatment. The aim of this study was to evaluate the effects in dogs with acute non-hemorrhagic diarrhea of the administration of an antibiotic combination in comparison to a nutraceutical product. Thirty dogs were enrolled and randomly assigned to two groups: 15 dogs (group A) received a nutraceutical commercial product while 15 dogs (group B) received an antimicrobial combination of metronidazole and spiramycin. For each dog, the Canine Acute Diarrhea Severity Index, the fecal microbiota and the Dysbiosis Index were assessed. Both stool consistency and frequency decreased on day 2 in the dogs of group A compared to baseline, while in group B, these parameters significantly decreased at days 3 and 4. The global concern for rising antibiotic resistance associated with indiscriminate use of antimicrobials, in both humans and animals, suggests the necessity of avoiding empirical and injudicious use of these molecules in diarrheic dogs. These results suggest that the nutraceutical treatment had a similar clinical effect compared to the antibiotic formulation, representing a valid antibiotic-sparing therapeutic approach in canine acute diarrhea.



Citation: Pignataro, G.; Di Prinzio, R.; Crisi, P.E.; Belà, B.; Fusaro, I.; Trevisan, C.; De Acetis, L.; Gramenzi, A. Comparison of the Therapeutic Effect of Treatment with Antibiotics or Nutraceuticals on Clinical Activity and the Fecal Microbiome of Dogs with Acute Diarrhea. *Animals* **2021**, *11*, 1484. <https://doi.org/10.3390/ani11061484>

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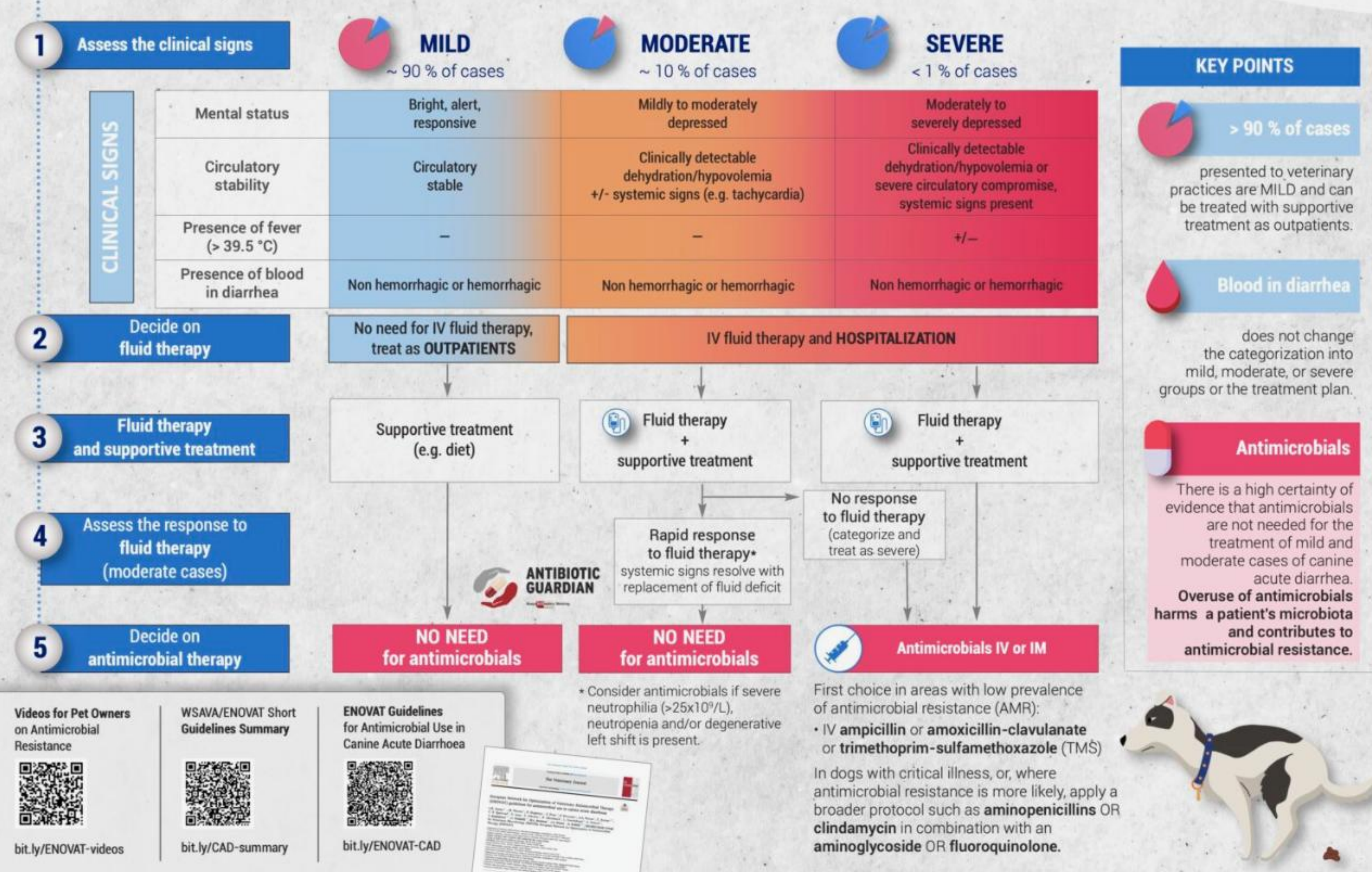
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Five Steps of Canine Acute Diarrhea Treatment

FIVE STEPS OF CANINE ACUTE DIARRHEA TREATMENT

Categorization and treatment plan is based on systemic clinical signs and not on the severity of gastrointestinal signs.



Chronic Diarrhea



Chronic Inflammatory Enteropathy

Diagnostic Criteria

- Chronic gastrointestinal signs (≥ 3 weeks)
- Histopathologic evidence of intestinal mucosal inflammation
- Exclusion of other underlying causes

Multifactorial pathophysiology

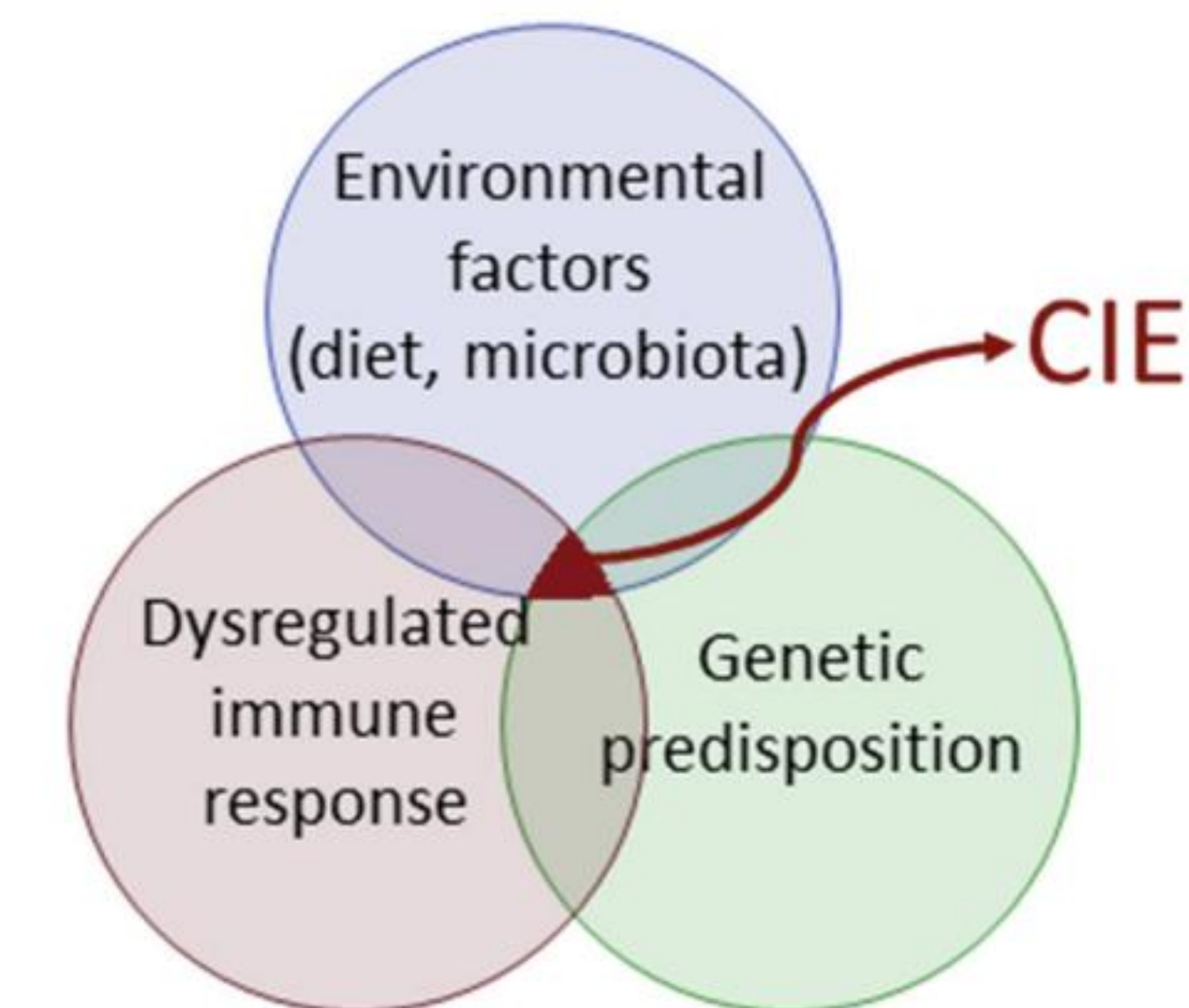


FIGURE 1
Complex disease pathogenesis in canine chronic inflammatory enteropathy (CIE).

Jergens and Heilmann. Frontiers in Veterinary Science 2022.



Chronic Inflammatory Enteropathy



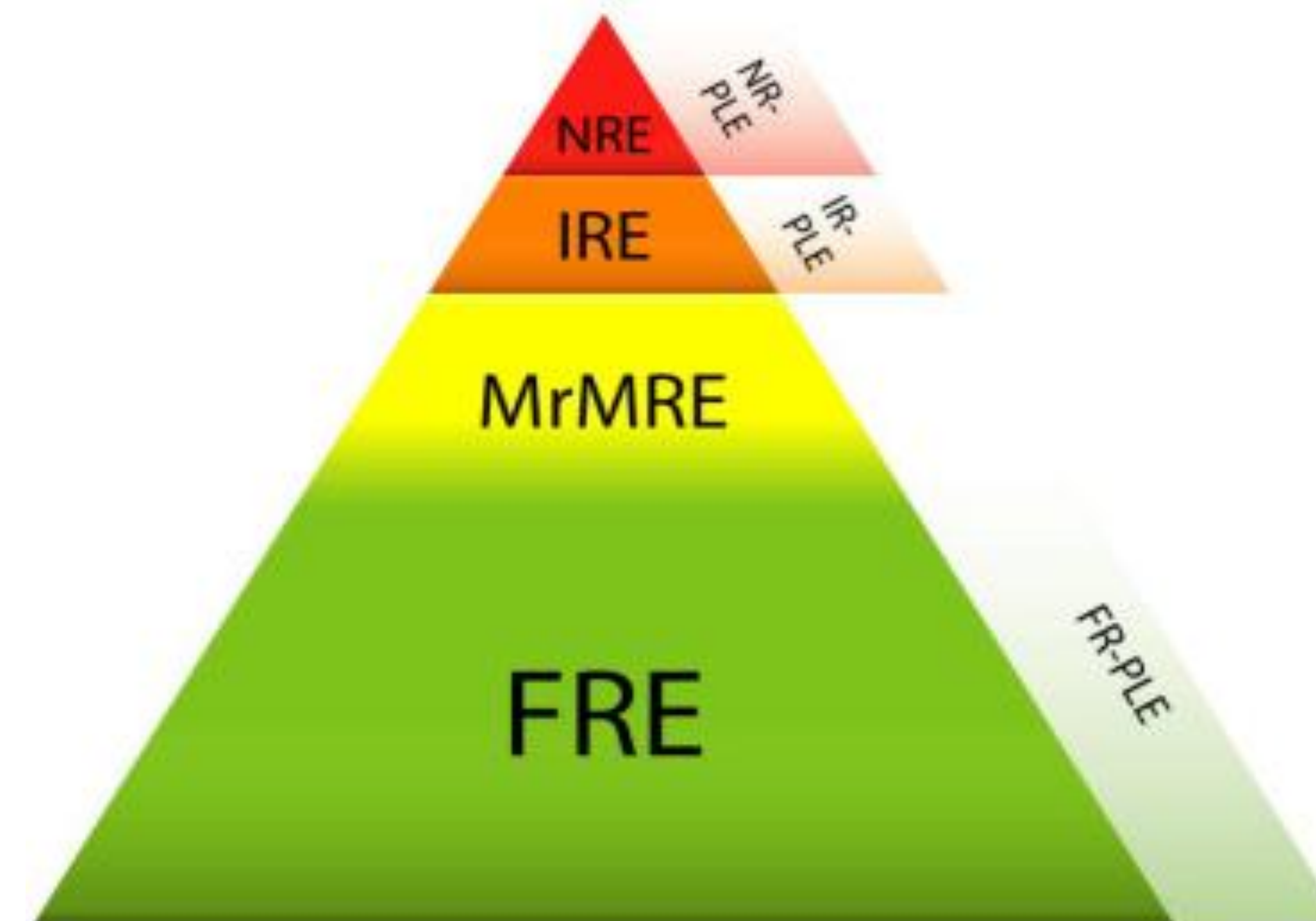
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Exclusion of other underlying causes

Multifactorial pathophysiology



Jergens and Heilmann. Frontiers in Veterinary Science 2022.
Dupouy-Manescau N, et al. Animals (Basel) 2024.



Initial visit/consultation

- Signalment and patient history (dietary history, deworming/vaccination status, medication, environment)
- Physical examination (clinical criteria of the CCECAI score documented)
- Blood testing (CBC, serum biochemistry panel incl. electrolytes, ± SBA) and urinalysis
- Fecal examination (fecal direct examination and flotation incl. *Giardia* spp. ELISA, ± other specific tests depending on geographic location, e.g. *Histoplasma* antigen ELISA)
- Gastrointestinal panel (serum cobalamin, folate, Spec cPL or validated DGGR lipase assay, cTLI, baseline cortisol)
- Diagnostic imaging (abdominal ultrasound, ± thoracic radiographs)

Initial elimination diet trial

- 2–4 weeks
- commercial novel antigen or hydrolyzed protein diet (alternative: home-cooked diet)

positive response → FRE → Follow-up
- visitation/consultation

no response

Second elimination diet trial

- 2–4 weeks, verify owner compliance
- commercial novel antigen or hydrolyzed protein diet (alternative: home-cooked diet)

positive response → FRE → Follow-up
- visitation/consultation

no response

Patient re-assessment

Further diagnostics

- Blood testing: recheck initial bloodwork for trends
- Gastrointestinal tissue biopsies
 - upper and lower gastrointestinal endoscopy: sampling of the stomach, duodenum/jejunum, ileum, and colon; laparoscopy or laparotomy: full-thickness biopsy of the stomach, duodenum, jejunum, ileum, ± mesenteric lymph nodes, ± liver biopsy, ± biopsy of other organs (if appropriate)
 - histopathological evaluation, ± immunohistochemistry, ± PARR, ± other diagnostics depending on suspected etiology (e.g., FISH analysis for adherent-and-invasive *E. coli*)

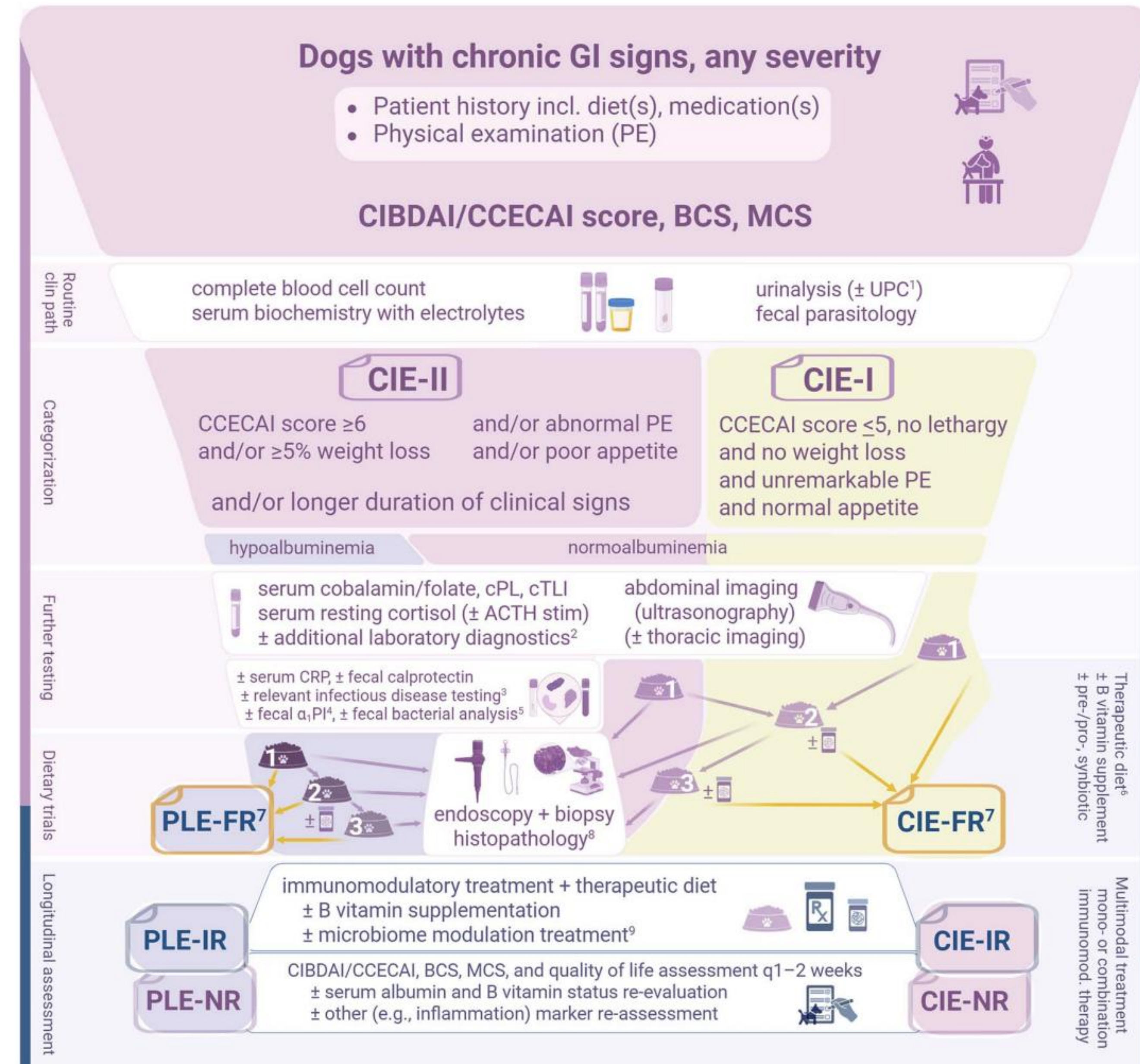
FIGURE 3
Diagnostic evaluation of dogs suspected with chronic inflammatory enteropathy (CIE). Selection and sequence of the individual diagnostic steps might vary and have to be tailored to the individual patient.

Jergens and Heilmann. Frontiers in Veterinary Science 2022.

Dogs with Chronic GI Signs, Any Severity?

Diagnosis

Monitoring



Heilmann RM. ACVIM Consensus Statement. 2026.



Choosing a Diet for Chronic GI Inflammation



Hydrolyzed



Novel Protein



Fibre Enhanced



Choosing a Diet for Chronic GI Inflammation



Hydrolyzed



Novel Protein



Digestibility



Fibre Enhanced

Key Point: Many diets overlap in their 'properties' which you can use to your advantage!



Fibre and Chronic Diarrhea



There are multiple studies investigating nutritional management of chronic diarrhea in dogs and cats with fiber-enhanced diets.

The majority of these studies are specific to animals with colitis signs, however some include animals with enteritis or enterocolitis clinical signs.

Simpson JM, et al. 1994.
Livet V, et al. 2013.
Royal Canin, et al. 2010.
Rossi G, et al. 2020.

Chronic studies indicate that the prognosis for dogs with fiber-responsive chronic gastrointestinal disease is good.

Long-term follow-up indicated that if fiber supplementation were removed or tapered, there was a chance that clinical signs would recur, indicating the need for long-term supplementation.

Dennis JS, et al. 1993.
Wernimont SM, et al. 2020.
Donadelli RA, et al. 2020.



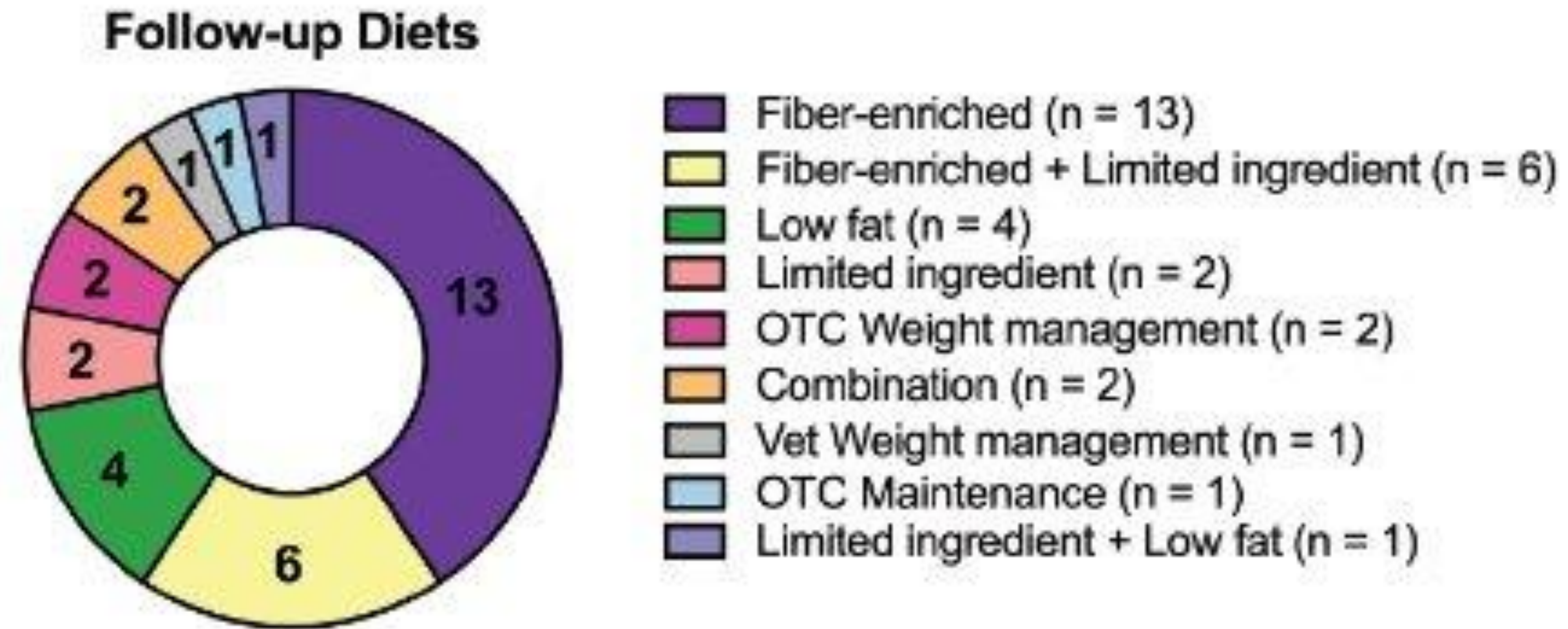
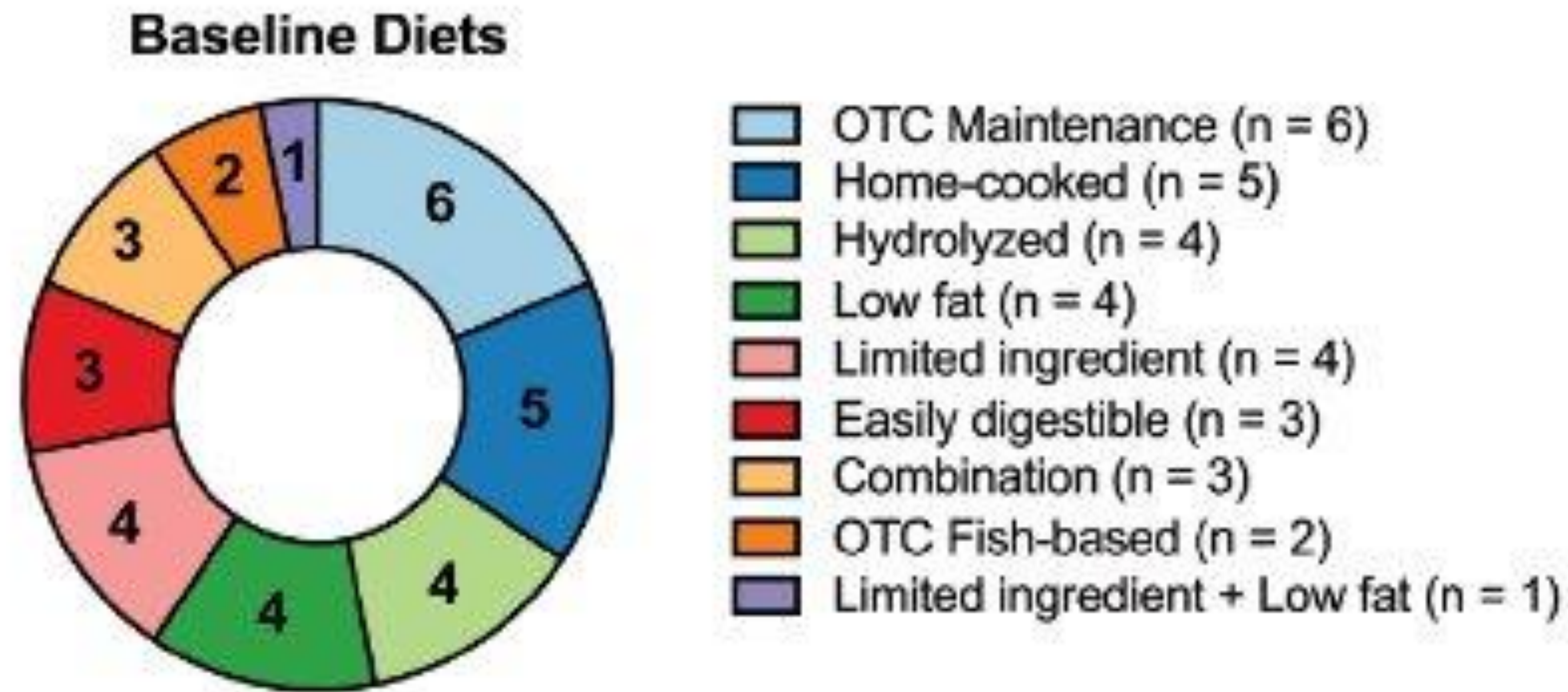


Figure 1—Diet types and number of dogs fed each diet type at baseline and follow-up. OTC = Over the counter.

Ford D, et al. 2026.



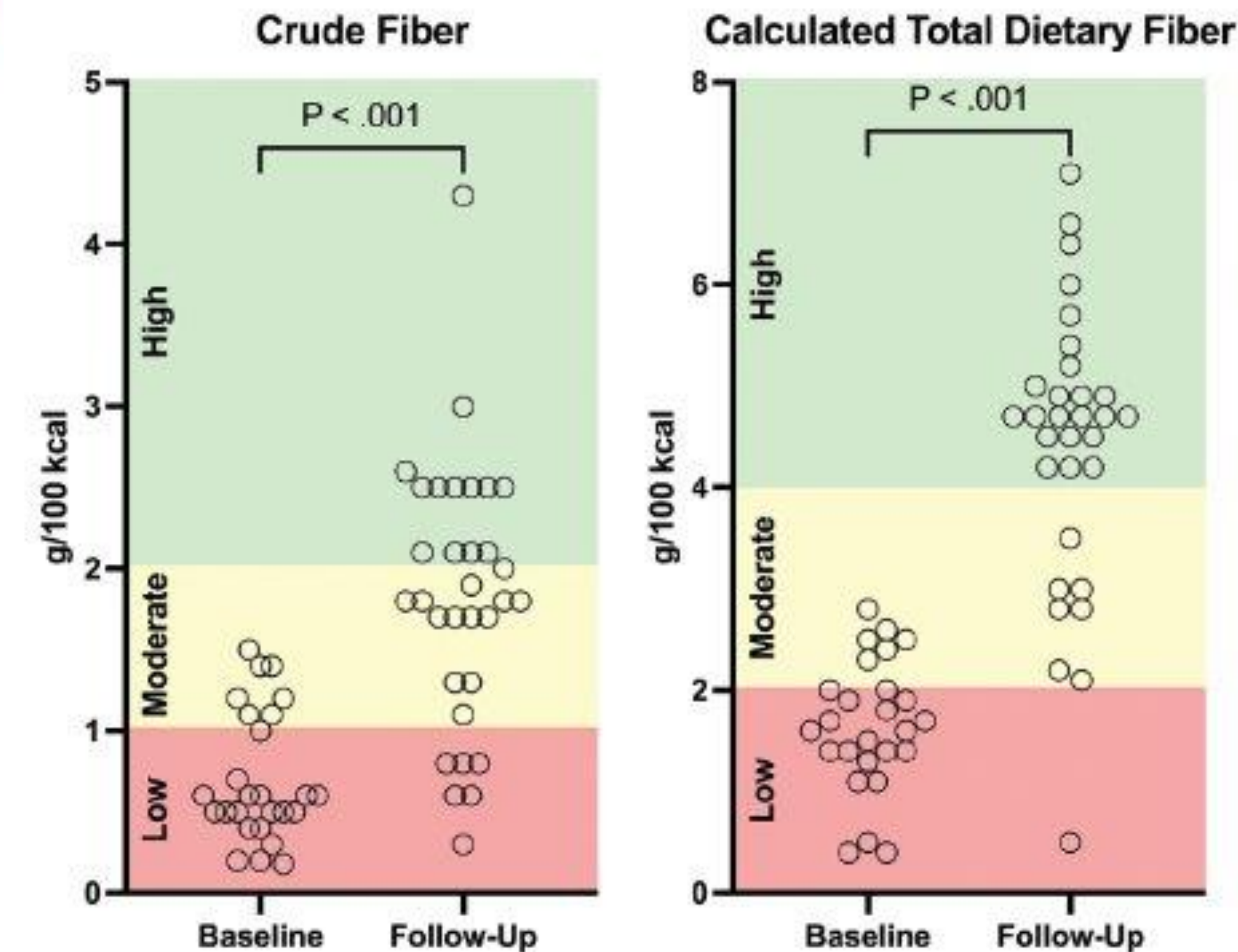


Figure 2—Dietary crude fiber and complete total dietary fiber (C-TDF) concentrations at baseline and follow-up on a g/100 kcal basis. Both crude fiber and complete total dietary fiber increased significantly ($P < .001$).

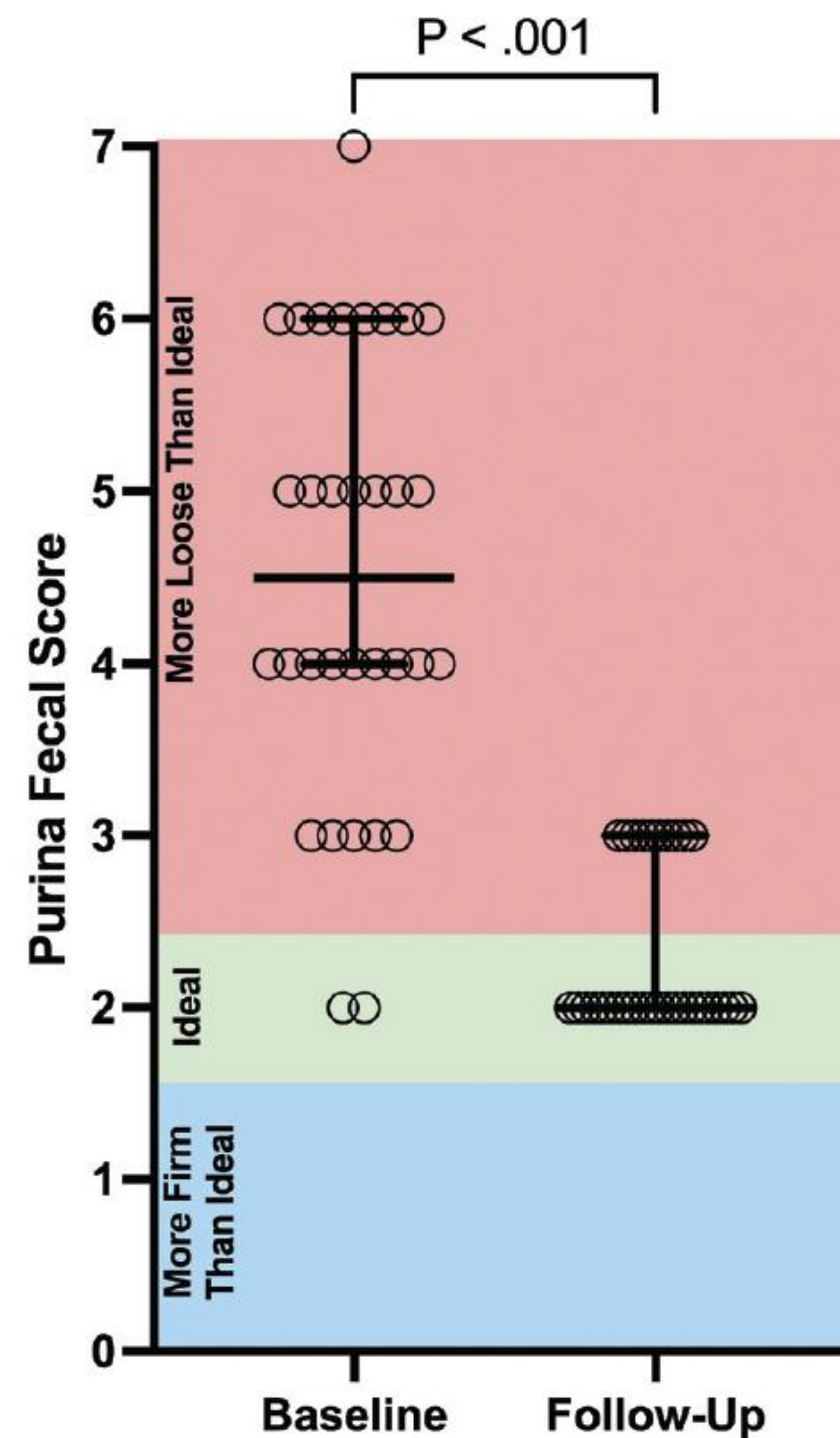


Figure 3—Stool quality, assessed with the Purina fecal score, significantly improved from baseline to follow-up ($P < .001$).

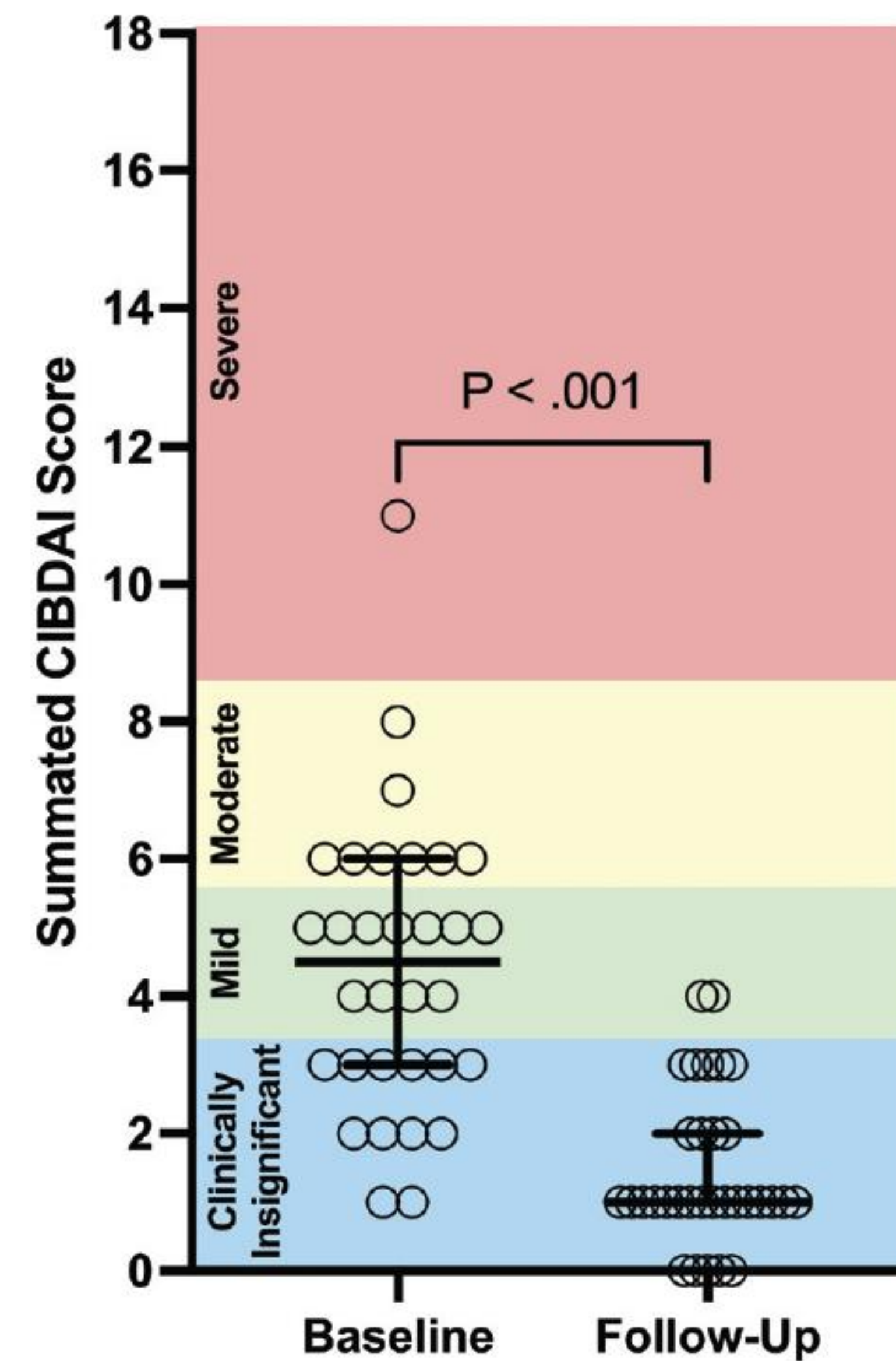


Figure 4—Canine inflammatory bowel disease activity index scores significantly improved from baseline to follow-up ($P < .001$).

Ford D, et al. 2026.



Pepperoni Case Study

- CBC/Chem/UA WNL
- T4 WNL
- KeyScreen Fecal PCR Negative
- Abdominal Ultrasound: SI muscularis thickening, mild chronic renal changes

Images

Figure 1: Stomach, essentially normal.

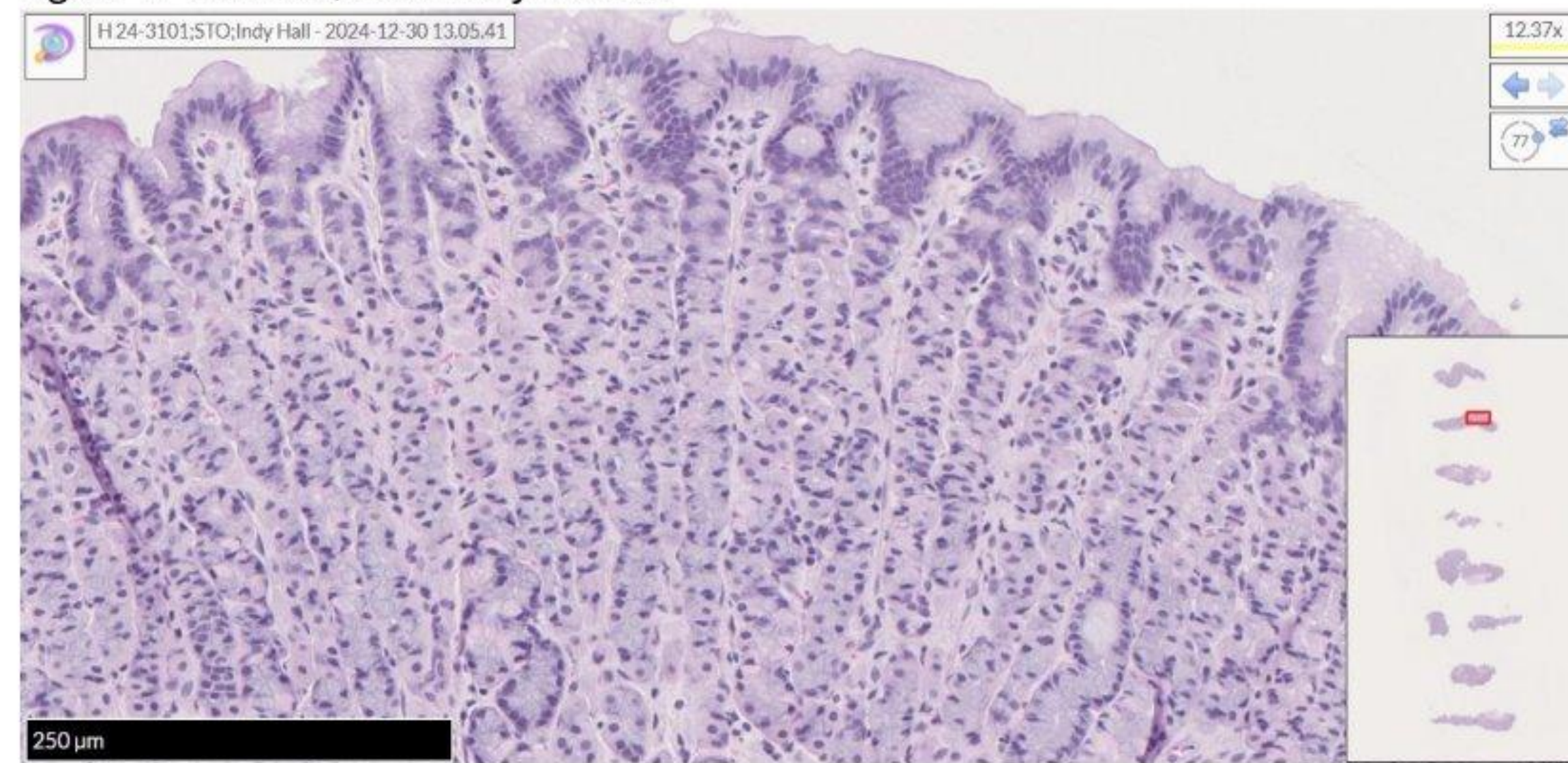


Figure 2: Duodenum with significant epithelial lymphocytes expanding through the epithelium, but otherwise fairly normal architecture.



Duodenum: 11 endoscopic sections with adequate depth and minimal crush artifact. There are evenly distributed lymphocytes in the epithelium and moderate numbers of mixed lymphocytes and plasma cells. The architecture is fairly normal with good villous length.

HISTOPATHOLOGICAL SCORING [MODIFIED WSAVA CRITERIA]

FEATURE	FUNDUS	ANTRUM	DUODENUM	ILEUM	COLON
Surface epithelium	0	0	0		
Gastric pit epithelium	0	0			
Villus stunting			0		
Crypt dilation/distortion			0		
Lacteal dilation			0		
Colonic goblet cells					
Mucosal fibrosis	0	0	0		
Intraepithelial lymphocytes	0	0	1-2		
LP lymphocytes and plasma	0	0	1-2		
LP eosinophils	0	0	0		
LP neutrophils	0	0	0		
LP macrophages	0	0	0		
Gastric lymphoid follicular	0	0			

0, normal; 1, mild; 2, moderate; 3, marked

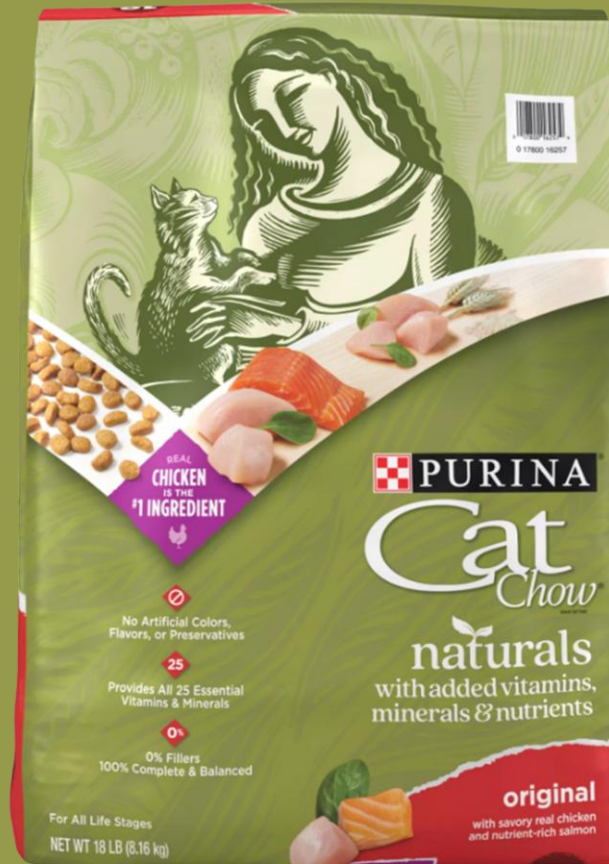
Colonic goblet cells: Inc2 (marked increase), Inc1 (mild increase), 0 (normal), Dec1 (mild decrease), Dec2 (marked decrease)



Pepperoni's Diet History

Purina Cat Chow Naturals Dry

3.71 kcal/gram
398 kcal/cup
Unknown TDF
0.4 g/100 kcal CF
4.1 g/100 kcal Fat



Hill's Prescription Diet z/d

3.84 kcal/gram
469 kcal/cup
0.4 g/100 kcal CF
1.2 g/100 kcal TDF
4.2 g/100 kcal Fat



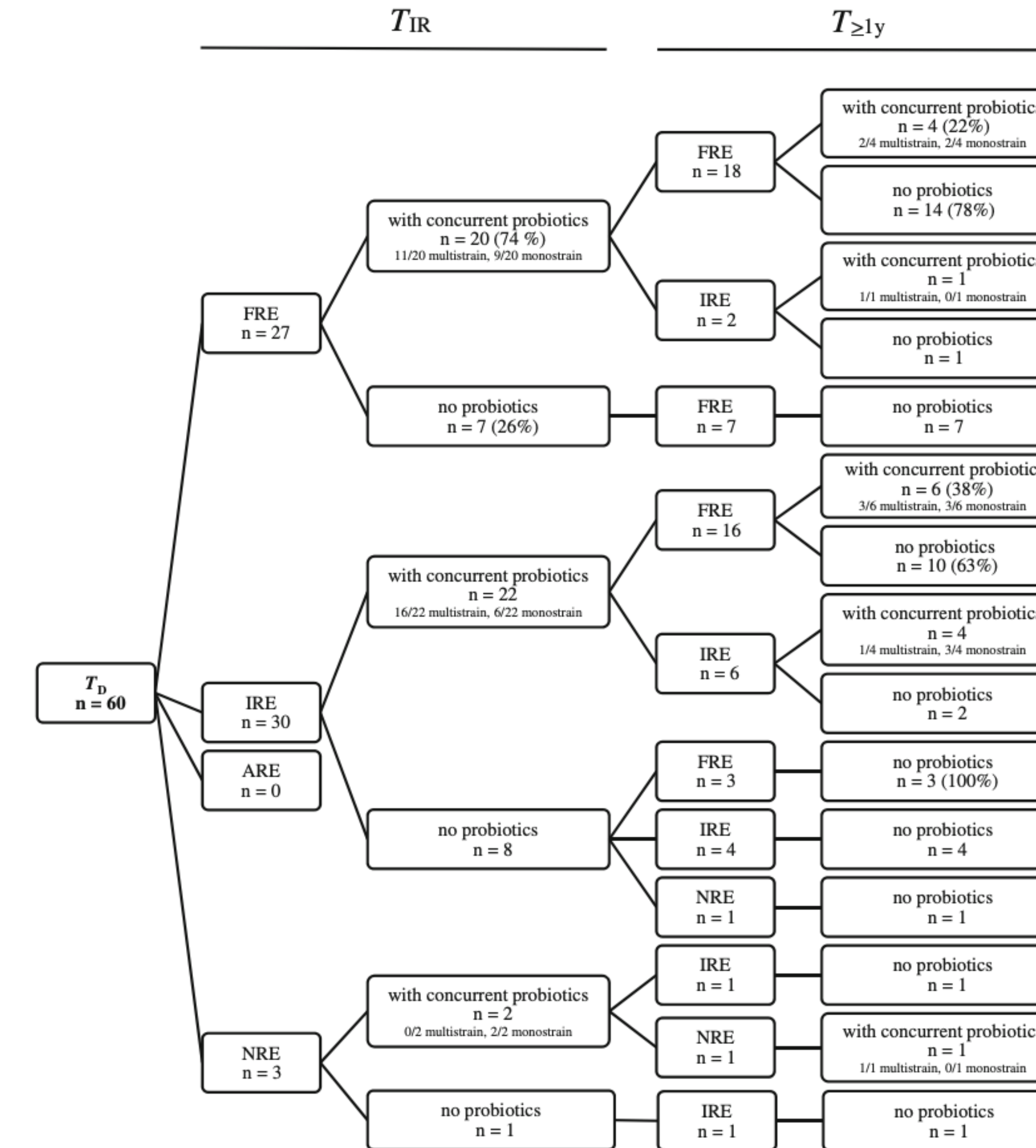
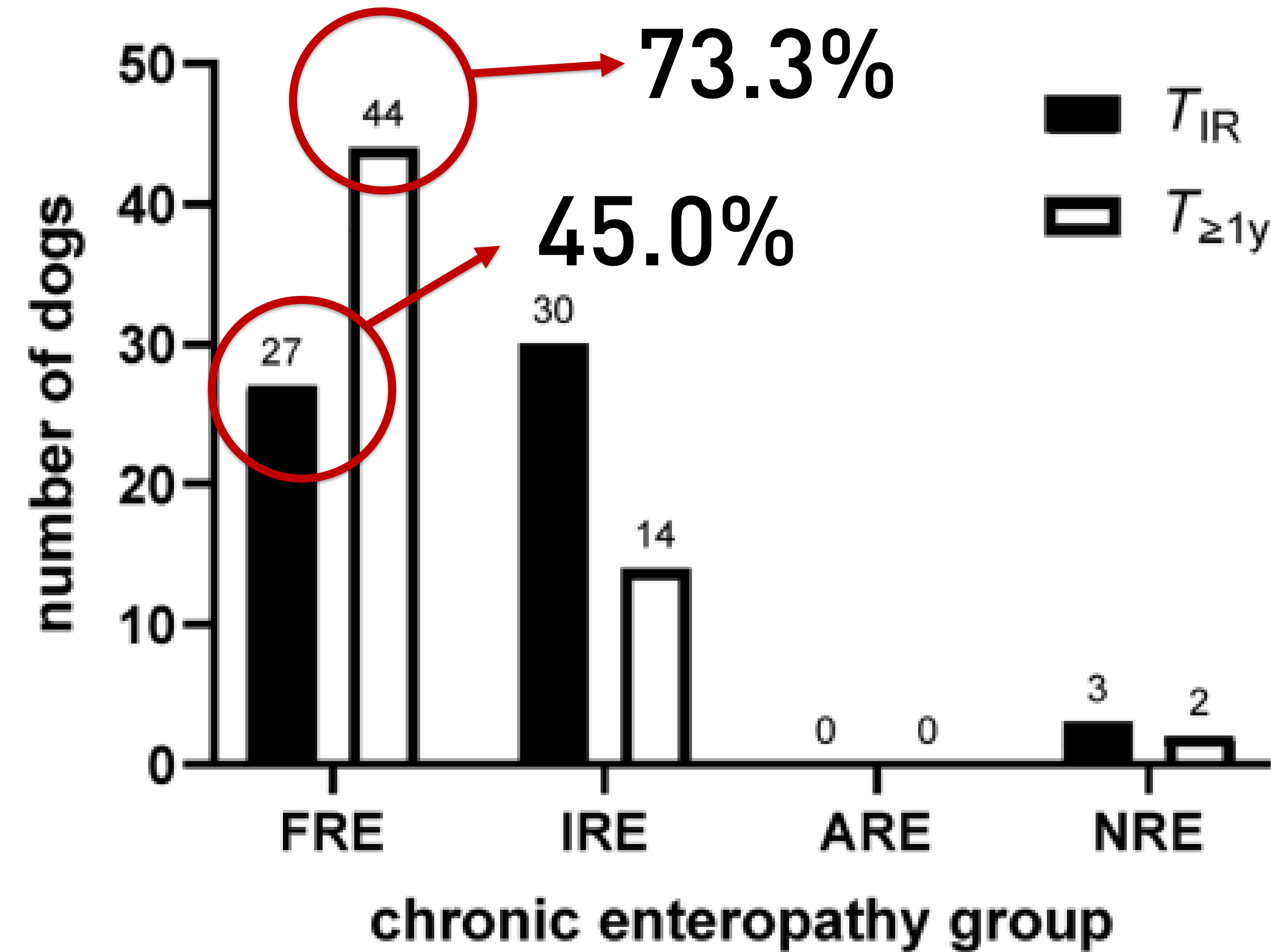
?

**Did we adequately explore
diet in Pepperoni's case?**





Play the Long Game: “Lane Switching”



Hodel S, et al. 2024.



Dedication Pays: Multiple Trials



Table 9. Available data on positive response to diet change at first re-check at our hospital (GE = Gastroenterology, IM = Internal Medicine).

Positive Response to Treatment	GE Dogs (<i>n</i> = 86)	IM Dogs (<i>n</i> = 88)
Only diet change	19/86 (22%)	21/88 (24%)
Diet change with probiotics	31/86 (36%)	5/88 (6%)
Diet change with corticosteroids	0/86 (0%)	4/88 (5%)
Diet change with antibiotics	0/86 (0%)	6/88 (7%)

3.9. Positive Response to Second and Third Diet Trial

Thirty-five of 92 (38%) GE dogs and 21/59 (24%) IM dogs were prescribed a second diet. The second diet was fed for a median of 2 weeks in both groups (range GE 1–24 weeks; IM 1–22 weeks), 23 GE dogs (66%), and 12 IM dogs (57%) responded positively to the second diet. Data on a third diet trial were available in 9/35 (26%) GE dogs and 7/21 (33%) IM dogs. The third diet was fed for a median of 4 weeks (range 2–84) in GE dogs and 3 weeks (range 1–5 weeks) in IM dogs, 4 GE dogs (44%), and 6 IM dogs (86%) responded positively to the third diet. None of the dogs with a positive response to a second and third diet change received additional new treatments. Results can be seen in Table 10.

Table 10. Available responses to second and third diet change documented at our hospital. A positive response was defined as any improvement in GI signs, whereas a negative response was defined as either no response or a worsening of signs. (GE = Gastroenterology, IM = Internal Medicine).

Outcome after Second Diet Change	GE Dogs (<i>n</i> = 35)	IM Dogs (<i>n</i> = 21)
Positive response	23/35 (66%)	12/21 (57%)
Negative response	7/35 (20%)	4/21 (19%)
No data	5/35 (14%)	5/21 (24%)
Outcome after Third Diet Change	GE Dogs (<i>n</i> = 9)	IM Dogs (<i>n</i> = 7)
Positive response	4/9 (44%)	6/7 (86%)
Negative response	1/9 (11%)	0/7 (0%)
No data	4/9 (44%)	1/7 (14%)

Schramm A, et al. 2022.

Diet History and Diet Selection are the Keys the Success!

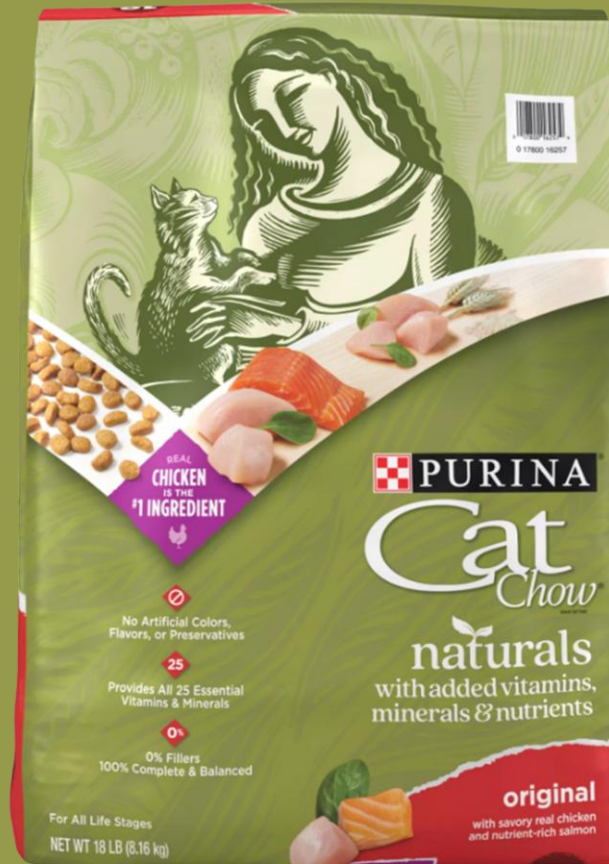


Pepperoni's Diet History



Purina Cat Chow Naturals Dry

3.71 kcal/gram
398 kcal/cup
Unknown TDF
0.4 g/100 kcal CF
4.1 g/100 kcal Fat



Hill's Prescription Diet z/d

3.84 kcal/gram
469 kcal/cup
0.4 g/100 kcal CF
1.2 g/100 kcal TDF
4.2 g/100 kcal Fat



?

What is the next best step?



Cat?

Vomiting?

Fat  Content

Digestibility

Hydrolyzed

Fiber Content

Novel Protein

Mixed Bowel Diarrhea?



Choosing a Diet



Fibre Enhanced



Novel Protein



Easily Digestible

KEY POINT: Many diets overlap in their 'properties' which you can use to your advantage



Pepperoni's Diet History



Purina Cat Chow Naturals Dry (whole Wheat)

3.71 kcal/gram
398 kcal/cup
Unknown TDF
0.4 g/100 kcal CF
4.1 g/100 kcal Fat



Hill's Prescription Diet z/d Beet pulp, citrus pulp, pecan shells

3.84 kcal/gram
469 kcal/cup
0.4 g/100 kcal CF
1.2 g/100 kcal TDF
4.2 g/100 kcal Fat



?

What is the next best step?



FELINE LIMITED INGREDIENT /HYDROLYZED DIETS

(In order of increasing phosphorus concentration)

Diet	Nutrients listed per 100 kcal								
	Kcal per can or cup	Kcal per gram	Phos (mg)	Prot (g)	Ca (mg)	Ca:P	Carb (g)	Fiber (g) (crude;TDF)	Fat* (g)
Royal Canin Renal Support + Hydrolyzed Protein (dry)	401	3.968	100	6.3	150	1.4	9.4	0.9; 2.3	4.9
Royal Canin Ultamino (dry)	335	3.762	128	6.3	153	1.2	10.7	0.9; 2.0	4.3
Hill's Prescription Diet z/d (dry)	469	3.847	144	8.4	184	1.3	10.0	0.4; 1.2	4.2
Royal Canin Hydrolyzed Protein Adult HP (dry)	330	3.923	148	6.3	172	1.2	9.4	1.0; 2.0	4.9
Royal Canin Gastrointestinal Hydrolyzed Protein (dry)	348	3.870	149	7.7	149	1.0	7.7	1.2; 2.6	5.0
Hill's Prescription Diet d/d Duck (5.5 oz)	178	1.142	167	8.6	173	1.0	4.5	1.6; 2.3	6.4
Royal Canin Urinary SO + Hydrolyzed Protein (dry)	334	3.625	170	7.1	190	1.1	10.0	1.4; 2.6	4.2
Hill's Prescription Diet d/d Duck & Green Pea Formula (dry)	486	3.897	174	8.2	181	1.0	8.2	1.0; 2.6	5.0
Hill's Prescription Diet d/d Venison & Green Pea Formula (dry)	486	3.893	178	8.2	191	1.1	8.2	1.0; 2.6	5.0
Rayne Rx TheraDiet Adult Health RSS Chunky Stew (6.4 oz)	182 /box	0.998	180	12.0	240	1.3	3.8	0.9; --	5.0
BLUE Natural Veterinary Diet HF (dry)	430	3.717	180	9.3	210	1.2	9.9	0.8; 1.8	4.1
Hill's Prescription Diet z/d (5.5 oz)	173	1.109	181	8.4	188	1.0	9.1	0.5; 1.4	4.6
Royal Canin Selected Protein Adult PR Loaf in Sauce (5.1 oz)	146	1.007	188	10.3	206	1.1	3.1	0.7; 1.4	5.6
Rayne Rx TheraDiet Kangaroo – Maint (freeze-dried)	203	4.320	190	13.0	190	1.0	1.6	0.6; --	5.8
Royal Canin Selected Protein Adult PD Loaf in Sauce (5.1 oz)	160	1.100	191	9.1	235	1.2	4.3	0.9; 2.1	5.9
Rayne Rx TheraDiet Adult Health RSS (dry)	383	3.576	200	9.1	240	1.2	10.4	0.7; --	3.7
Royal Canin Satiety + Hydrolyzed Protein (dry)	238	3.016	210	10.7	240	1.1	9.2	4.6; 7.4	3.0
Farmina VetLife Derma HF (dry)	353	3.529	230	7.9	280	1.2	7.9	0.5; --	3.1
Royal Canin Selected Protein Adult PR (dry)	334	3.548	230	8.5	250	1.1	10.1	1.0; 2.6	3.5
BLUE Natural Veterinary Diet HF (5.5 oz)	161	1.030	250	11.6	280	1.1	10.7	0.2; 1.2	2.6
Rayne Rx TheraDiet Kangaroo – Maint Chunky Stew (6.4 oz)	160 /box	0.881	250	13.7	260	1.0	4.0	0.2; --	4.5
nine+line Allergy Care Kangaroo (5.3 oz)	44 /oz	1.553	256	8.4	436	1.7	0.6	--; 0.1	7.5
Purina Pro Plan Veterinary Diets HA (dry)	291	3.428	260	10.3	370	1.4	10.0	0.6; 1.6	3.3
Royal Canin Selected Protein Adult PD (dry)	332	3.531	267	8.5	307	1.2	10.0	1.0; 2.6	3.5
BLUE Natural Veterinary Diet NP (dry)	414	3.703	270	8.5	430	1.6	9.2	0.6; 1.2	4.5
BLUE Natural Veterinary Diet NP (5.5 oz)	204	1.310	280	7.6	420	1.5	1.9	0.2; 0.5	7.6
Farmina VetLife Derma Management w/ Pork (3 oz)	68	0.798	288	8.2	313	1.1	9.8	0.6; --	4.4
Rayne Growth/Sensitive GI Chunky Stew (6.4 oz)	192 /box	1.053	300	12.3	330	1.1	3.6	0.2; --	5.3
Rayne Kangaroo Maint Paté (5.5 oz)	145	0.927	340	13.4	470	1.4	4.8	0.4; --	4.3
AAFCO minimum (adult maintenance)			125	6.5	150	<i>n/a</i>	<i>n/a</i>	<i>n/a</i>	2.3

*Fat is rarely a primary nutrient of concern for cats! You *may* consider it for a cat with severe acute pancreatitis.



FELINE HIGH FIBER DIETS (not all-inclusive list – see additional options on reduced calorie & urinary diet lists)
(In order of decreasing TDF)

Diet	Kcal per cup/can	Kcal per gram	Fiber (g) (crude; TDF)	Nutrients listed per 100 kcal			
				Prot (g)	Phos (mg)	Ca (mg)	Ca:P
Royal Canin Kitten Spayed/Neutered (dry)	280	3.260	3.0; 5.5	9.9	280	310	1.1
Royal Canin Appetite Control (dry)	284	3.302	2.5; 5.5	9.8	290	320	1.1
Hill's Science Diet Adult Hairball Control Savory Chicken Entrée (5.5 oz)	179	1.147	3.0; 4.9	7.5	201	213	1.1
Hill's Science Diet Adult Indoor Salmon & Vegetable Medley (5.5 oz)	124	0.792	2.7; 4.7	10.0	202	254	1.3
Hill's Science Diet Adult Multiple Benefit (dry)	271	3.295	2.5; 4.5	9.7	185	223	1.2
Hill's Science Diet Adult 7+ Hairball Control (dry)	421	3.713	2.3; 4.5	8.1	172	198	1.2
Hill's Science Diet Adult Hairball Control Light (dry)	279	3.180	2.4; 4.4	11.6	229	284	1.2
Hill's Science Diet Indoor Adult (dry)	327	3.492	2.3; 4.4	9.7	207	248	1.2
Hill's Science Diet Adult Hairball Control (dry)	321	3.651	2.3; 4.3	8.8	204	232	1.1
Hill's Science Diet Adult Urinary Hairball Control Savory Chicken (5.5 oz)	159	1.019	2.3; 4.2	9.1	190	259	1.4
Hill's Science Diet Adult 7+ Indoor (dry)	321	3.574	2.2; 4.2	9.0	171	210	1.2
Hill's Science Diet Adult Urinary Hairball Control (dry)	418	3.689	2.3; 4.1	8.8	181	207	1.2
Hill's Prescription Diet Multi-Organ (dry)	468	3.840	2.1; 4.1	9.3	140	222	1.6
Royal Canin Hairball Care (dry)	342	3.564	1.7; 4.0	9.1	270	310	1.1
Hill's Science Diet Indoor Adult 11+ (dry)	396	3.772	2.1; 3.8	8.5	169	212	1.3
Hill's Science Diet Adult Indoor Ocean Fish Entrée (5.5 oz)	174	1.115	2.1; 3.6	9.5	165	178	1.1
Hill's Science Diet Adult Light Liver & Chicken Entrée (5.5 oz)	147	0.943	1.9; 3.5	9.7	200	243	1.2
Hill's Science Diet Adult Light (dry)	288	3.171	1.7; 3.5	11.0	224	281	1.3
Hill's Science Diet Adult Oral Care (dry)	326	3.715	2.2; 3.4	8.3	168	208	1.2
Royal Canin American Shorthair Adult (dry)	333	3.656	1.3; 3.4	8.1	200	230	1.2
Royal Canin Aging Spayed/Neutered 12+ (dry)	347	3.687	1.3; 3.4	7.8	160	190	1.2
Hill's Prescription Diet t/d (dry)	269	3.520	2.2; 3.3	9.1	206	261	1.3
Royal Canin Indoor 7+ (dry)	308	3.587	0.9; 3.3	7.7	200	240	1.2
Hill's Prescription Diet GI Biome Stress Chicken & Veg Stew (2.9 oz)	68	0.826	1.2; 3.2	9.3	159	238	1.5
Royal Canin Urinary Care (dry)	317	3.520	1.2; 3.2	8.8	200	240	1.2
Royal Canin Dental Care (dry)	311	3.536	1.4; 3.2	8.1	260	290	1.1
Purina Pro Plan Veterinary Diets EN Savory Selects With Chicken (5.5 oz)	149	0.956	0.7; 3.1	14.4	260	340	1.3
Hill's Prescription Diet Multi-Organ (5.5 oz)	171	1.098	1.8; 3.1	9.7	131	186	1.4
Royal Canin Gastrointestinal Moderate Calorie (dry)	313	3.436	1.6; 3.1	9.6	270	300	1.1
Royal Canin Indoor Adult (dry)	325	3.534	1.0; 3.1	7.8	230	300	1.3
Royal Canin Indoor Long Hair (dry)	381	3.596	1.2; 3.0	8.4	290	290	1.0
Royal Canin Maine Coon Adult (dry)	362	3.846	1.3; 3.0	7.8	210	250	1.2
Purina Pro Plan Veterinary Diets DH (dry)	279	3.426	1.3; 2.9	10.2	360	410	1.1
Royal Canin Fit & Active (dry)	334	3.666	1.0; 2.9	8.8	280	300	1.1
Royal Canin Gastrointestinal Fiber Response (dry)	362	3.690	0.7; 2.9	8.0	260	280	1.1
Hill's Prescription Diet Gastrointestinal Biome Stress (dry)	421	3.807	0.8; 2.8	9.2	178	185	1.1
Royal Canin Ragdoll Adult (dry)	402	3.795	0.9; 2.8	8.0	200	240	1.2
Royal Canin Hair & Skin Care (dry)	430	3.984	1.1; 2.8	8.0	220	270	1.2
Royal Canin Gastrointestinal Fiber Response (3 oz)	61	0.712	0.8; 2.1	9.4	230	270	1.2
AAFCO minimum (adult maintenance)			n/a	6.5	125	150	n/a



Pepperoni's Diet History



Purina Cat Chow Naturals Dry

(whole Wheat)

3.71 kcal/gram
398 kcal/cup
Unknown TDF
0.4 g/100 kcal CF
4.1 g/100 kcal Fat



Hill's Prescription Diet z/d

Beet pulp, citrus pulp, pecan shells

3.84 kcal/gram
469 kcal/cup
0.4 g/100 kcal CF
1.2 g/100 kcal TDF
4.2 g/100 kcal Fat



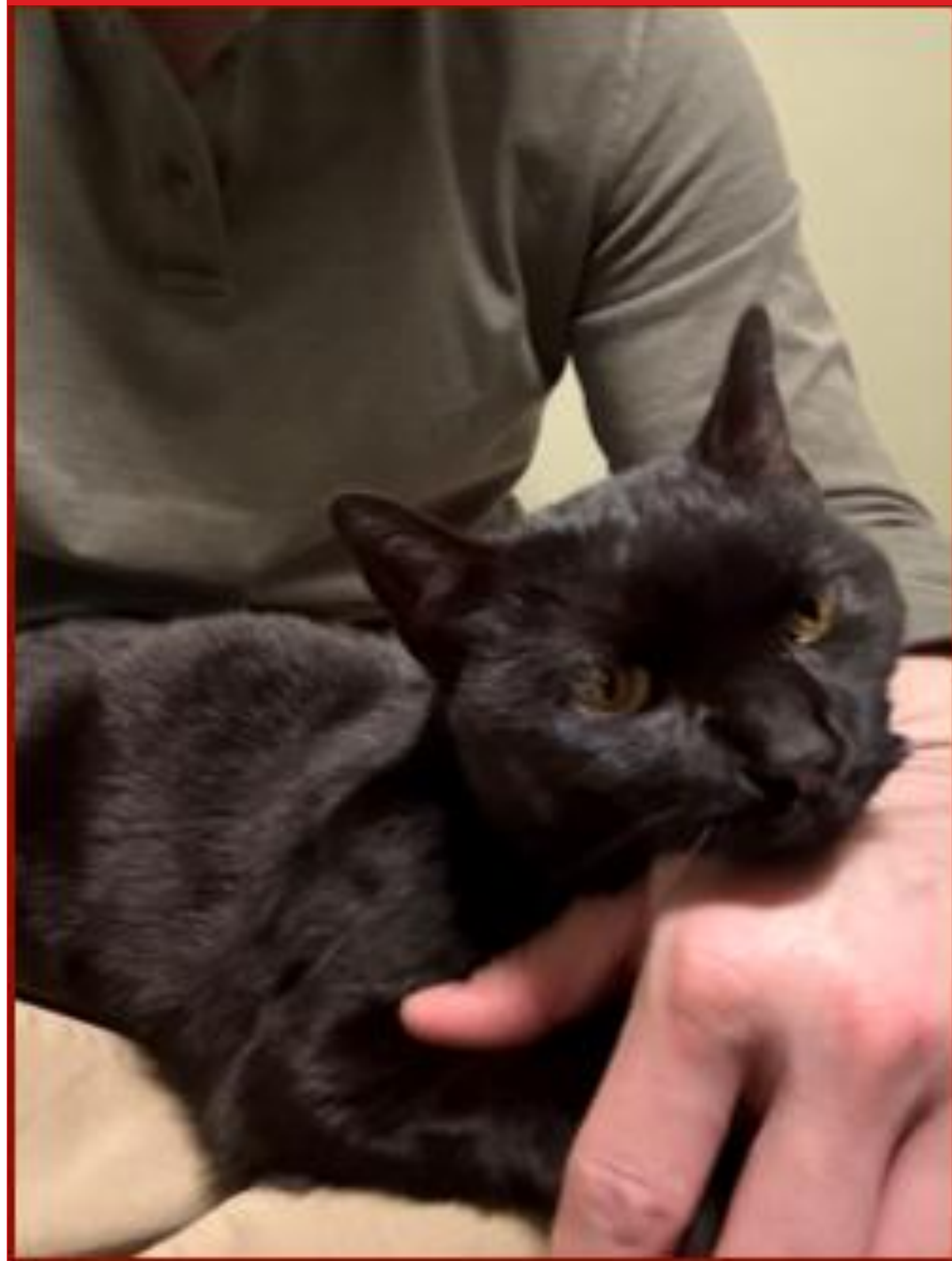
Royal Canin GI Fibre Response

(Psyllium, Chichory Root, FOS)

3.69 kcal/gram
362 kcal/cup
0.7 g/100 kcal CF
2.9 g/100 kcal TDF
3.5 g/100 kcal Fat



Clinical Update



Evaluation

Vomiting (3 time in one month)

▫ 95% reduction

Mixed Bowel Diarrhea Resolved

Weight stable, eating normally





Irritable Bowel Syndrome

The role of stress?



IBS is Common in People!

Functional gastrointestinal disorders account for 40% of all referrals to gastroenterologists

IBS is the most common

- Greatly impacts quality of life
- Estimated that 12 % of patients worldwide have IBS
- Young patients
- Female



IBS Subtypes

IBS subtype based on the patient's predominant bowel habits on days with abnormal bowel movements

- IBS with constipation (IBS-C)
- IBS with diarrhea (IBS-D)
- IBS with mixed symptoms (IBS-M)
- Unsubtyped (IBS-U)



Table 1. Rome IV Criteria for the Irritable Bowel Syndrome.*

Patient has recurrent abdominal pain (≥ 1 day per week, on average, in the previous 3 mo), with an onset ≥ 6 mo before diagnosis

Abdominal pain is associated with at least two of the following three symptoms:

Pain related to defecation

Change in frequency of stool

Change in form (appearance) of stool

Patient has none of the following warning signs:

Age ≥ 50 yr, no previous colon cancer screening, and presence of symptoms

Recent change in bowel habit

Evidence of overt GI bleeding (i.e., melena or hematochezia)

Nocturnal pain or passage of stools

Unintentional weight loss

Family history of colorectal cancer or inflammatory bowel disease

Palpable abdominal mass or lymphadenopathy

Evidence of iron-deficiency anemia on blood testing

Positive test for fecal occult blood

* The information is from Mearin et al.¹ GI denotes gastrointestinal.

N Engl J Med. 2017 Jun
29;376(26):2566-2578.

Diagnosis is a Challenge

The Rome IV criteria are widely used for diagnosing IBS. These criteria include:

- Recurrent abdominal pain or discomfort for at least 6 months.
- Pain that improves with defecation or is associated with a change in bowel habits.
- Two or more of the following symptoms:
 - Change in stool frequency
 - Change in stool form
 - Bloating
 - Sensation of incomplete evacuation



IBS AT A GLANCE		
CAUSES	SYMPTOMS	TREATMENT
 STRESS, ANXIETY, OR DEPRESSION	 CONSTIPATION AND/OR DIARRHEA	 AVOID TRIGGER FOODS
 CHANGE IN THE GUT'S MICROBIOTA	 ABDOMINAL PAIN	 REGULAR EXERCISE
 GENETICS	 GAS AND/OR BLOATING	 MEDICATION

New York Presbyterian. "Health Matters". 2022.

Pandora Syndrome

"A name like 'Pandora' syndrome seems appropriate for at least 2 reasons," says Buffington. "First, it does not identify any specific cause or organ, and second, it seems to capture the dismay and dispute associated with the identification of so many problems (evils) outside the organ of interest of any particular subspecialty."

-Tony Buffington-



The Real 'FIC'

Healthy and FIC cat study

- **Sickness behaviors increase in both populations with stress and changes in environment**

- Sickness behaviors included:

- Vomiting

- Diarrhea

- Anorexia

- Others...

- **Standardized feeding, litter box management and play time decreased abnormal signs by 75-80%**

Stella JL, et al. 2011.

Fibre and IBS

There are two studies supporting the use of fiber-enhanced diets in both non-inflammatory and idiopathic chronic colitis.

Response documented to both soluble and insoluble fiber sources.

Implementing fiber supplementation allowed adjunctive medical treatments to be tapered or discontinued in some dogs.

Leib MS, et al. 2000.

Lecoindre P, et al. 2011.



Irritable Bowel Syndrome (IBS)

Diagnosis of exclusion and included behavioral evaluation

Fiber is a key dietary focus

Treatment should be focused on reducing stress, behavior modification, environmental enrichment, and pain management

In cats, focus on litter box, feeding regimen (diet, type, flavor), safe space, owner interactions, and environment

Striking Balance

Test heavily for concurrent disease (if possible)

- Interpret testing wisely

Support the ecosystem

- Nutrition, Fiber, Probiotics

Manage gut-brain axis

- Behavior medications, behavior modifications

Set expectations

- Some damage can't be undone





Constipation



Fibre and Constipation



The basis of the recommendation for soluble fiber is its ability to form gels which facilitate fecal passage and induce prokinetic effect.

It has also been recommended to use insoluble or non-fermentable fibers as these are more likely to increase fecal bulk and dilute luminal toxins.

Data is limited in cats and very scant in dogs.

Dimski DS, et al. 1991.
Dimski DS, et al. 1994.
Barry KA, et al. 2010.
Hallman JER, et al. 1996.

Strompfova V, et al. 2013.
Oh HGA, et al. 2011.
Frieche V, et al. 2011.



Nutritional Factors



Water

- Maintain hydration status
- Adjust delivery of water
 - ☐ Clean water
 - ☐ Available water
 - ☐ Moving water
 - ☐ Flavor water (bouillon/broth)
- Moist/wet food

Fiber

Digestibility

Caloric Density

Water

Fiber

- Increase fecal water
- Increase colonic motility
- Improved intestinal transit

Digestibility

Caloric Density



Constipation

the infrequent or difficult evacuation of feces (does not imply a permanent loss of function)

Obstipation

intractable constipation that has become refractory to cure or control



Megacolon Cats



Motility patterns are nearly completely abolished in megacolon cats

Fiber supplementation is contraindicated in these patients

Minimize dietary fiber and use low-residue/easily digestible diets



SUMMARY

Fiber is a complex component of canine and feline nutrition.

Understanding the basics of fiber is useful to practicing clinicians.

Canine and feline gastrointestinal disease can often be effectively managed with fiber-enhanced diets and in many cases offers many advantages over alternative treatment options.

Specific types or amounts of fiber for specific disease management is limited.





THANK YOU

